

Which Member Carries:

Two-State Pairs from Atoms to Gas Sensors, and Döbereiner's Triads as Balanced Pairs

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Abstract

Which member of a pair carries (the charge, the population, the change of nuclear charge, the electron of a sensor surface, the centre of a triad) is decided in chemistry by one law, the two-state pair $p = 1/(1 + e^\varepsilon)$ split into a common part $\frac{1}{2}$ and a difference $\delta = -\frac{1}{2} \tanh(\varepsilon/2)$. We show that an observation of which member carries determines δ and nothing else, and we evaluate the pair in six settings. Potassium carries the charge of the separated pair Na, K at every temperature: the minority configuration $\text{Na}^+ + \text{K}$ has occupancy 3.87×10^{-14} at 300 K as a single pair and 1.97×10^{-7} in an equimolar gas, where mass action halves the exponent. The mean nuclear charge change per decay of ^{40}K is $+0.7856(22)$ from the 2017 ENSDF evaluation and $+0.7918(10)$ from the 2023 KDK branching ratios, 2.6 combined standard uncertainties apart. The upper fine-structure level of aluminium holds the majority above $T_{1/2} = 232.61$ K. We prove that a code shared by the absent rows of two channels turns their correlation exactly into $(r + fd_x d_y)/\sqrt{(1 + fd_x^2)(1 + fd_y^2)}$, whose sign flips when the geometric-mean code distance crosses $\sqrt{-r/f}$; from the present-row moments of two sensor channels of the UCI air-quality record this closed form reproduces the coded correlations $+0.086931$ (code -200 , geometric-mean distance 4.9133) and -0.075630 (code 0 , distance 4.0595) to 10^{-16} .

First-order mass action makes the ionosorbed-oxygen occupancy of a metal-oxide surface an exact logistic in $\varepsilon = \ln(\Phi_+/\Phi_-)$, the log-ratio of the electron-release and electron-trapping fluxes. Fitted to the five sensors of the same record on the common CO axis, the two-state law is preferred to the offset power law for every sensor ($\Delta\text{AIC} = 11.7$ to 209.0), with slopes $\beta = 1.18$ to 1.45 , and it orders the sensors into a ladder of half-points: S3 (1.68 mg m^{-3} , the falling tungsten-oxide surface, which reads the opposite member) $<$ S5 (4.61) $<$ S2 (5.40) $<$ S1 (7.49 , at the upper edge of the data) $<$ S4 (not identified), with $\text{S3} < \text{S5} < \text{S2}$ in both halves of the year. Humidity enters every sensor on the side of the reducing gas. In this urban record NO_2 rises with CO, so the oxidizing member is not separable through NO_2 ; the opposite-sign pair is CO against NO on S4, with balance slope 3.28 . The record's benzene column is an exact power law of the titania sensor, $1.23978 \times 10^{-4} (S_2 - 324.785)^{1.743158}$ to 1.6×10^{-15} : a calibration output, not a measurement. From the Madelung order we prove the period-pair law $L(p) = 2[(p+1)/2]^2$ and that a vertical triad is exact, $Z_2 = \frac{1}{2}(Z_1 + Z_3)$, if and only if the two periods it spans have equal length. Of the 54 vertical triads, 27 are exact, including all four of Döbereiner's; their middle member is the zero of the pair $(-s, +s)$, $s \in \{8, 18, 32\}$. The same mass-action pair decides which gas survives the NH_4SH deck of the giant planets. The proofs solve the steady state of first-order mass action, decompose pooled moments and count subshells in the Madelung order; the sensor results are least-squares fits with day-block bootstrap intervals.

1 Introduction

Döbereiner observed in 1829 that chemically similar elements come in threes whose middle atomic weight is close to the mean of the other two [1]. Scerri recast such triads in terms of

atomic number, where the middle value can be the exact mean, and used them to compare the medium-long and left-step forms of the periodic table [2, 3, 4]; the lengths of the periods follow the Madelung $(n + \ell, n)$ order, whose origin Allen and Knight discuss [5]. The branching of ^{40}K between β^- decay and electron capture has two recent determinations, the 2017 ENSDF evaluation [6] and the 2023 KDK measurement of ground-state capture [7, 8]. For metal-oxide gas sensors the ionosorption model [9, 10] and the depletion theory of Yamazoe and Shimano [11] account for the empirical power laws, and the air-quality record of De Vito et al. [12, 13], a year of hourly readings of five such sensors beside reference analysers, is a standard record for their field calibration. For tables with absent values, the attenuation of correlations by imputation is known qualitatively [14, 15].

We treat all of these as one pair. A pair of states with weights $(p, 1 - p)$ can be written without loss as a common part and a difference,

$$p = \frac{1}{2} + \delta, \quad 1 - p = \frac{1}{2} - \delta, \quad \delta = -\frac{1}{2} \tanh(\varepsilon/2), \quad \varepsilon = \ln \frac{1 - p}{p}. \quad (1)$$

Indeed $\varepsilon = \ln((1 - p)/p)$ inverts to $p = 1/(1 + e^\varepsilon)$, and then $p - \frac{1}{2} = (1 - e^\varepsilon)/(2(1 + e^\varepsilon)) = -\frac{1}{2} \tanh(\varepsilon/2)$. The log-odds $\pm\varepsilon$ form a pair whose zero $\varepsilon = 0$ is located in turn at a temperature, a code distance, a gas concentration and a middle atomic number; independent adsorption sites and independent products of a charge transfer combine without changing the law; an exact vertical triad of the periodic table is $Z_2 + \{-s, 0, s\}$; and an observation of which member carries determines δ and nothing else, since every factor common to the two members cancels, while in a group the shared end of two exact triads closes the first and opens the next.

Companion papers. The same law with $\varepsilon = -z(V - V_{1/2})$ is the gate law of excitable membranes [17]: there the half-point is a voltage, and for a gate moving an effective charge through the membrane field ε is again a free-energy difference in units of $k_B T$. The same mass-action pair decides the top ice deck of the giant planets [18] (Section 7.11). An exact triad $Z_2 + s\{-1, 0, 1\}$ is the digit set of balanced ternary scaled by s ; the arithmetic and radix economy of that digit set are treated in [19]. The synthesis paper [16] places these results beside those of the other fields.

1.1 Main results

Our main results are the following.

- (1) *The pair and its half-point* (Section 2). Every two-state distribution is the pair (1), whose difference δ is odd under the exchange $\varepsilon \mapsto -\varepsilon$ of the members. A pair with degeneracies g_l, g_u and gap $\Delta E > 0$ balances at a positive temperature if and only if $g_u > g_l$, and then exactly at $T_{1/2} = \Delta E / (k_B \ln(g_u/g_l))$ (Theorem 1).
- (2) *Sodium and potassium* (Section 3). One valence electron shared between a sodium and a potassium core is carried by potassium at every temperature: the configurations $\text{Na} + \text{K}^+$ and $\text{Na}^+ + \text{K}$ lie 0.7984132(3) eV apart with degeneracy ratio 1, so balance is unreachable. The minority configuration has occupancy 3.87×10^{-14} at 300 K as a single pair and 1.97×10^{-7} in an equimolar gas, where the two products form independently and mass action halves the exponent.
- (3) *The two branches of ^{40}K* (Section 4). The mean nuclear charge change per decay is the difference of the pair, $\langle \Delta Z \rangle = \tanh(\varepsilon/2) = 2p_{\beta^-} - 1$. It is +0.7856(22) from the 2017 ENSDF evaluation and +0.7918(10) from the 2023 KDK branching ratios; the two differ by 0.0062, that is by 2.6σ , with $\sigma = 0.0024$ the combined standard uncertainty.
- (4) *The fine-structure doublet of aluminium* (Section 5). The upper level $^2P_{3/2}^\circ$ ($g = 4$, 112.061 cm $^{-1}$) holds the majority above $T_{1/2} = 232.61$ K and tends to $\frac{2}{3}$.

- (5) *Joint absences* (Section 6). If two channels with present-row correlation r are absent together on a fraction f of the rows and these rows carry a shared code at standardised distances d_x , d_y from the present-row means, their correlation is exactly

$$r_{\text{coded}} = \frac{r + f d_x d_y}{\sqrt{(1 + f d_x^2)(1 + f d_y^2)}};$$

for $r < 0$ and $d_x d_y > 0$ its sign flips when the geometric-mean distance $\sqrt{d_x d_y}$ crosses $\sqrt{-r/f}$, however unequal d_x and d_y are, and one coded correlation bounds how far its code lies from the data (Theorem 2). From the present-row moments of the sensors PT08.S1 and PT08.S3 of the UCI air-quality record this closed form reproduces exactly the coded correlations of the record, +0.086931 for the code -200 ($\sqrt{d_x d_y} = 4.9133$) and -0.075630 for the code 0 (4.0595), to 10^{-16} ; the flip lies between them, at 4.4424.

- (6) *The two-state law of a sensor surface, and the sensor ladder* (Section 7). First-order mass action makes the ionosorbed-oxygen occupancy of a metal-oxide grain the exact logistic $\theta = 1/(1 + e^\varepsilon)$ with $\varepsilon = \ln(\Phi_+/\Phi_-)$, the log-ratio of the electron-release and electron-trapping fluxes (Theorem 5). Fitted to the five sensors of the air-quality record on the common CO axis, the two-state law is preferred to the offset power law for every sensor ($\Delta\text{AIC} = 11.7$ to 209.0), with slopes $\beta = 1.18$ to 1.45 near first-order mass action, and its half-points form the ladder

$$\text{S3 (1.68)} < \text{S5 (4.61)} < \text{S2 (5.40)} \quad \text{mg m}^{-3} \text{ of CO,}$$

identified inside the data and ordered in the same way in both halves of the year, with S1 (7.49) at the upper edge of the data and S4 not identified. The tungsten-oxide sensor S3 falls with every traffic gas: it reads the opposite member of the same pair. Humidity enters every sensor on the side of the reducing gas. In this urban record NO_2 rises with CO, so the CO and NO_2 coefficients carry the same sign on S1, S2, S3 and S5 and the oxidizing member is not separable through NO_2 ; the opposite-sign pair appears as CO against NO on S4, with balance slope 3.28 [2.90, 3.73].

- (7) *The benzene column* (Section 7.9). The column $\text{C}_6\text{H}_6(\text{GT})$ of the record is the exact power law $1.23978 \times 10^{-4} (S_2 - 324.785)^{1.743158}$ of the titania sensor S2 on all 8991 rows, to a relative deviation of 1.6×10^{-15} . It is a calibration output, not an independent measurement, and its form is the power-law regime of the two-state law.
- (8) *The period-pair law and Döbereiner's triads* (Sections 8 to 10). Period p has the capacity of the Madelung group $n + \ell = p + 1$, so $L(p) = 2[(p + 1)/2]^2$ (Theorem 6). A vertical triad is exact, $Z_2 = \frac{1}{2}(Z_1 + Z_3)$, if and only if the two periods it spans have equal length (Theorem 9). Of the 54 vertical triads, 27 are exact; their middle member is the zero of the pair $(-s, +s)$, $s \in \{8, 18, 32\}$. Döbereiner's four triads are all exact, Li–Na–K with $s = 8$ and Ca–Sr–Ba, Cl–Br–I and S–Se–Te with $s = 18$; their middle atomic weights lie 0.13 to 1.57% below the mean of the outer pair, and in all six exact triads of groups 1, 2, 16 and 17 the middle first ionization energy lies strictly between the outer two.
- (9) *The giant planets* (Section 7.11). The mass-action pair of the sensor surface also decides the top ice deck of the giant planets: NH_3 and H_2S are titrated 1:1 by the NH_4SH lattice, their difference $x_N - x_S = 2\chi \sinh(\varepsilon/2)$ is conserved, and the surviving vapour is $\text{sign}(N - S)$, so the zero $N/S = 1$ separates Jupiter (4.73) and Saturn (3.16) from Uranus (0.195) and Neptune (0.170) [18].

1.2 Outline of the method

The exact statements come from first-order mass action and from counting. Theorems 1 and 5 solve the steady state of a two-state balance and read off its zero. Theorem 2 decomposes the

pooled moments of a coded table into present-row moments and code distances, and its bound is the Cauchy–Schwarz inequality; Theorem 4 does the same for mean imputation. Theorem 6 counts the subshells of each Madelung group, and Theorem 9 compares the gaps down a group with the period lengths through an alignment lemma (Theorem 7). The atomic and nuclear results evaluate these formulas on tabulated data. The sensor results fit the two-state law to the hourly record by nonlinear least squares, compare it with the offset power law through the Akaike criterion, and take 95% intervals from a bootstrap that resamples whole days, so that the serial correlation of hourly data is respected (Section 7.3).

1.3 Organisation and notation

Section 2 treats the pair and its half-point, Sections 3 to 5 the three evaluated atomic and nuclear pairs, Section 6 the coding of absent measurements, Section 7 the sensor ladder, the benzene column and the giant planets, Sections 8 to 10 the periodic table, and Section 11 horizontal triads and lanthanide-contraction pairs; Section 12 concludes, and Section 13 lists every input. Throughout, p is the occupancy of a member, ε its log-odds against the other and $\delta = p - \frac{1}{2}$ the difference; $T_{1/2}$, d^* and $C_{1/2}$ are the zeros $\varepsilon = 0$ in temperature, code distance and concentration; k_B is the Boltzmann constant. For the periodic table, $L(p)$ and $\Sigma(p)$ are the length and the last atomic number of period p , and $\eta = x_2/\bar{x} - 1$ is the deviation of a middle member from the mean $\bar{x}$ of the outer pair. Numbers in square brackets after an estimate are 95% day-block bootstrap intervals.

2 The pair and its half-point

In the pair (1) the common part $\frac{1}{2}$ says nothing about which member is favoured; the difference δ says everything, and it is odd under $\varepsilon \mapsto -\varepsilon$, the exchange of the two members. For a two-state system in thermal equilibrium ε is the free-energy difference in units of $k_B T$; for competing decay branches it is the logarithm of a ratio of partial decay constants (Section 4); for a sensor surface it is the logarithm of the ratio of the electron-release and electron-trapping fluxes (Section 7.2).

If the two members have degeneracies g_l , g_u and energy separation $\Delta E > 0$, the upper member’s occupancy is

$$p_u(T) = \frac{g_u e^{-\Delta E/k_B T}}{g_l + g_u e^{-\Delta E/k_B T}}, \quad \varepsilon(T) = \frac{\Delta E}{k_B T} - \ln \frac{g_u}{g_l}.$$

Proposition 1. *The pair is balanced ($p_u = \frac{1}{2}$) at a positive temperature iff $g_u > g_l$, and then exactly at $T_{1/2} = \Delta E/(k_B \ln(g_u/g_l))$; as $T \rightarrow \infty$, $p_u \rightarrow g_u/(g_u + g_l)$. If $g_u \leq g_l$ the upper member remains the minority at every temperature. Equivalently, the upper member is the majority at temperature T iff $\Delta E < k_B T \ln(g_u/g_l)$.*

Proof. Dividing numerator and denominator by $g_u e^{-\Delta E/k_B T}$ gives $p_u = 1/(1 + e^{\varepsilon(T)})$, so $p_u > \frac{1}{2}$ iff $\varepsilon < 0$ and $p_u = \frac{1}{2}$ iff $\varepsilon = 0$. Since $\Delta E > 0$, $\varepsilon(T)$ decreases strictly and continuously from $+\infty$ (as $T \rightarrow 0^+$) to $-\ln(g_u/g_l)$ (as $T \rightarrow \infty$). If $g_u > g_l$ the limit is negative and ε has exactly one zero, $\Delta E/k_B T = \ln(g_u/g_l)$, i.e. $T = T_{1/2}$; if $g_u \leq g_l$ the limit is ≥ 0 and $\varepsilon > 0$ at every finite T . As $T \rightarrow \infty$, $p_u \rightarrow 1/(1 + g_l/g_u) = g_u/(g_u + g_l)$. The last statement is $\varepsilon(T) < 0$ rewritten. $\square$

3 One electron, two cores: sodium and potassium

Consider isolated gas-phase atoms. One valence electron shared between a sodium and a potassium core at large separation (the two lowest dissociation limits of NaK⁺) is in one of the configurations Na + K⁺ or Na⁺ + K, separated by the difference of first ionisation energies,

$$\Delta E = 5.13907696(25) - 4.34066373(9) = 0.7984132(3) \text{ eV} \quad [20, 21],$$

with the lower configuration Na + K⁺: potassium carries the charge. (The uncertainty is the quadrature sum of the two stated uncertainties.) Both neutral atoms are ²S_{1/2} and both ions ¹S₀ [21], so each configuration has electronic degeneracy 2 · 1 = 1 · 2 = 2, the degeneracy ratio is 1, and by Theorem 1 balance is unreachable at every temperature. As a two-state system, $\varepsilon = \Delta E/k_B T$ with $k_B = 8.617333262 \times 10^{-5} \text{ eV K}^{-1}$ [22], and the minority occupancy is $1/(1 + e^\varepsilon)$:

$T \text{ (K)}$	300	310	1000
$\varepsilon = \Delta E/k_B T$	30.884	29.888	9.265
occupancy of Na ⁺ + K	3.87×10^{-14}	1.05×10^{-13}	9.47×10^{-5}

Solving $1/(1 + e^\varepsilon) = p$ for the temperature gives $T = \Delta E/(k_B \ln((1 - p)/p))$: the minority configuration reaches 10^{-6} at 671 K and 10^{-3} at 1341 K.

A pair is not a mixture. In a gas prepared with equal amounts c_0 of Na and K⁺, consider the charge-transfer equilibrium $\text{Na} + \text{K}^+ \rightleftharpoons \text{Na}^+ + \text{K}$. For ideal gases the equilibrium constant is the ratio of the partition functions per unit volume, each with energies measured from its own ground state, $K_{\text{eq}} = (q_{\text{Na}^+} q_{\text{K}})/(q_{\text{Na}} q_{\text{K}^+}) e^{-\Delta E/k_B T}$. The electronic factors cancel (degeneracy 1 · 2 on each side), and excited electronic states do not contribute below 1000 K: the lowest ones, K $4p^2 P_{1/2,3/2}^\circ$ at 12985.19 and 13042.90 cm⁻¹ (1.61 eV) [20], hold a fraction 2.2×10^{-8} of K atoms at 1000 K, and all other excitations lie higher. The translational factors, proportional to $m^{3/2}$, leave $[(1 - m_e/m_{\text{Na}})(1 + m_e/m_{\text{K}^+})]^{3/2} = 1 - 1.5 \times 10^{-5}$, with $m_e = 5.486 \times 10^{-4} \text{ u}$ [22] and $m \approx 23 \text{ u}, 39 \text{ u}$; we set this factor to 1. With converted fraction x the amounts are $x c_0$ of Na⁺ and of K and $(1 - x)c_0$ of Na and of K⁺, so $K_{\text{eq}} = x^2/(1 - x)^2$ and

$$\frac{x}{1 - x} = e^{-\Delta E/2k_B T}, \quad x = 1.97 \times 10^{-7} \text{ (300 K)}, \quad 9.64 \times 10^{-3} \text{ (1000 K)}.$$

The exponent is halved because the two products form independently: the single pair and the mixture answer different questions, and the square root is the whole difference between them.

4 The two branches of ⁴⁰K

⁴⁰K (0.0117% of natural potassium [23]) decays by β^- to ⁴⁰Ca ($\Delta Z = +1$) or by electron capture (and a 10^{-5} fraction of β^+) to ⁴⁰Ar ($\Delta Z = -1$). Here $\varepsilon = \ln(\lambda_{\beta^-}/\lambda_{\text{EC}})$ is the logarithm of a ratio of partial decay constants, not a thermal quantity. The branching fractions are $p_{\beta^-} = \lambda_{\beta^-}/(\lambda_{\beta^-} + \lambda_{\text{EC}}) = 1/(1 + e^{-\varepsilon})$ and $p_{\text{EC}} = 1 - p_{\beta^-}$, where p_{EC} stands for electron capture together with the β^+ branch (the column $p_{\text{EC}} + p_{\beta^+}$ of the table below), which also lowers Z by one. The net nuclear charge change per decay is the difference of the pair,

$$\langle \Delta Z \rangle = (+1)p_{\beta^-} + (-1)p_{\text{EC}} = \frac{1 - e^{-\varepsilon}}{1 + e^{-\varepsilon}} = \tanh(\varepsilon/2) = 2p_{\beta^-} - 1,$$

whose standard uncertainty is twice that of p_{β^-} , since the two fractions sum to one. The entropy per decay is $-p_{\beta^-} \log_2 p_{\beta^-} - p_{\text{EC}} \log_2 p_{\text{EC}}$.

evaluation	p_{β^-} (%)	$p_{\text{EC}} + p_{\beta^+}$ (%)	$\langle \Delta Z \rangle$	entropy (bits/decay)
ENSDF 2017 [6]	89.28(11)	10.72(11)	+0.7856(22)	0.4914
KDK 2023 [7, 8]	89.59(5)	10.41	+0.7918(10)	0.4818

The KDK entry combines their measured ground-state capture 0.098(25)% (statistical and systematic uncertainties 0.023 and 0.010 combined in quadrature) with the capture to the excited state, 10.31(4)%, both from their re-evaluation of the decay scheme using the partial decay constants of [8], and the β^+ branch 0.001% of [6]; the three electron-capture-side entries and p_{β^-} sum to 0.99999 and are normalised to one before $\langle \Delta Z \rangle$ and the entropy are formed. The two values of $\langle \Delta Z \rangle$ differ by 0.0062, or 2.6 combined standard uncertainties ($\sqrt{0.0022^2 + 0.0010^2} = 0.0024$). The common part $\frac{1}{2}$ counts decays; the difference is the charge. A balanced branching would carry no net charge on average and one full bit per decay.

5 The fine-structure doublet of aluminium

The ground term of neutral aluminium is $3s^23p^2P^\circ$, with $^2P^\circ_{1/2}$ ($g = 2$) lowest and $^2P^\circ_{3/2}$ ($g = 4$) at 112.061 cm^{-1} [20], i.e. $\Delta E/k_B = hc\tilde{\nu}/k_B = 161.231 \text{ K}$ (13.894 meV) with the second radiation constant $hc/k_B = 1.438776877 \text{ cm K}$ [22]; the next level, $3s^24s^2S_{1/2}$, lies at $25347.756 \text{ cm}^{-1}$ [20]. For atoms thermalised at temperature T within the ground-term manifold (for example in a buffer gas), $p_u(T) = 2e^{-\Theta/T}/(1+2e^{-\Theta/T})$ with $\Theta = 161.231 \text{ K}$, and Theorem 1 with $g_u/g_l = 2$ gives

$$T_{1/2} = \frac{hc\tilde{\nu}}{k_B \ln 2} = 232.61 \text{ K},$$

above which the upper level holds the majority, tending to $\frac{2}{3}$. Values of p_u : 0.198 (77 K), 0.406 (150 K), $\frac{1}{2}$ (232.61 K), 0.539 (300 K), 0.630 (1000 K), 0.666 (10^5 K). By the last clause of Theorem 1, a 2P ground term with the $J = \frac{1}{2}$ level lowest has its excited level in the majority at 300 K iff the splitting is below $k_B \cdot 300 \text{ K} \cdot \ln 2 = 144.5 \text{ cm}^{-1}$; aluminium's 112.061 cm^{-1} qualifies. At $T_{1/2}$ each level holds $\frac{1}{2}$, so each of the two lower states has probability $\frac{1}{4}$ and each of the four upper states $\frac{1}{8}$; the six ground-term states then carry $2 \cdot \frac{1}{4} \log_2 4 + 4 \cdot \frac{1}{8} \log_2 8 = 1 + 1.5 = 2.5$ bits of entropy (the maximum, $\log_2 6 = 2.585$ bits, is approached only as $T \rightarrow \infty$, where all six states have probability $\frac{1}{6}$).

6 The coding of absent measurements

Two channels X, Y are recorded on n rows. On the n_{11} rows where both are present they have means μ_x, μ_y , variances $v_x, v_y > 0$ and covariance c , all with divisor n_{11} , and $r = c/\sqrt{v_x v_y}$. Suppose first that the other $n - n_{11} = fn$ rows, $0 < f < 1$, are absent in *both* channels and are filled with constants (a_x, a_y) . Write the standardised code distances

$$d_x = \frac{a_x - \mu_x}{\sqrt{v_x}}, \quad d_y = \frac{a_y - \mu_y}{\sqrt{v_y}}.$$

Proposition 2 (joint absences). *The correlation over all n rows is*

$$r_{\text{coded}} = \frac{r + f d_x d_y}{\sqrt{(1 + f d_x^2)(1 + f d_y^2)}}. \quad (2)$$

(a) (Mean-coding is invisible.) If $d_x = d_y = 0$ then $r_{\text{coded}} = r$, the value obtained by deleting the absent rows.

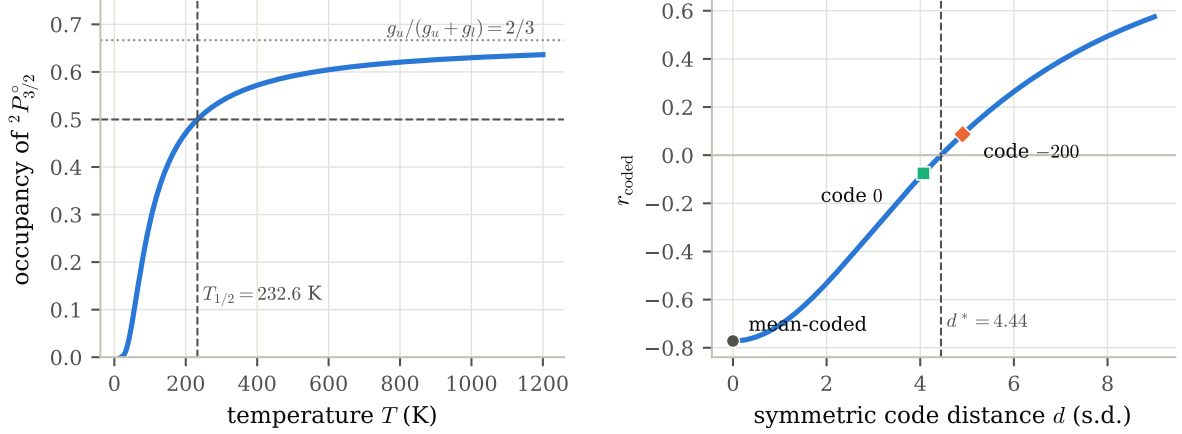


Figure 1: Left: occupancy of Al $^2P_{3/2}^o$ for atoms thermalised in the ground term; the pair balances at $T_{1/2} = 232.6$ K and tends to $g_u/(g_u + g_l) = \frac{2}{3}$. Right: the symmetric-code curve $G(d) = (r + fd^2)/(1 + fd^2)$ of Theorem 2 for $r = -0.771918$ and $f = 366/9357$ (channels PT08.S1 and PT08.S3 of the air-quality record); the sign flips at $d^* = 4.44$. The markers place the coded correlations $+0.0869$ (code -200) and -0.0756 (code 0) at their symmetric-equivalent distances $d_{\text{sym}} = 4.904$ and 4.068 . The directly computed geometric-mean distances of the two codes, 4.913 and 4.060 , lie on the sides that Theorem 2(d) requires: above d_{sym} for -200 and below it for 0 .

(b) (*Sign.*) r_{coded} has the sign of $r + f d_x d_y$. If $r < 0$ and $d_x d_y > 0$ the sign changes exactly when the geometric-mean distance $\sqrt{d_x d_y}$ crosses $d^* = \sqrt{-r/f}$, however unequal d_x and d_y are.

(c) (*A shared distant code manufactures agreement.*) As $|d_x|, |d_y| \rightarrow \infty$, $r_{\text{coded}} \rightarrow \text{sign}(d_x d_y)$, whatever r was.

(d) (*Bound.*) $|r_{\text{coded}}| \leq |r + f d_x d_y|/(1 + f|d_x d_y|)$, with equality iff $|d_x| = |d_y|$. Consequently, if $d_x d_y > 0$ and $G(d) = (r + fd^2)/(1 + fd^2)$ is the symmetric-code curve, an observed r_{coded} defines $d_{\text{sym}} \geq 0$ by $G(d_{\text{sym}}) = r_{\text{coded}}$, i.e. $d_{\text{sym}}^2 = (r_{\text{coded}} - r)/(f(1 - r_{\text{coded}}))$, and

$$\sqrt{d_x d_y} \geq d_{\text{sym}} \text{ if } r_{\text{coded}} > 0, \quad \sqrt{d_x d_y} \leq d_{\text{sym}} \text{ if } r \leq r_{\text{coded}} < 0.$$

Proof. Write $\alpha_x = a_x - \mu_x$, $\alpha_y = a_y - \mu_y$. The pooled mean of X is $m_x = (1 - f)\mu_x + f a_x = \mu_x + f \alpha_x$, so a present row deviates from it by $(x_i - \mu_x) - f \alpha_x$ and a coded row by $(1 - f)\alpha_x$. Summing squares over the $n_{11} = (1 - f)n$ present rows (the cross term vanishes because $\sum (x_i - \mu_x) = 0$) and the fn coded rows, and dividing by n , the pooled variance is

$$(1 - f)(v_x + f^2 \alpha_x^2) + f(1 - f)^2 \alpha_x^2 = (1 - f)(v_x + f \alpha_x^2) = (1 - f)v_x(1 + f d_x^2);$$

likewise for Y . In the same way the pooled covariance is $(1 - f)(c + f^2 \alpha_x \alpha_y) + f(1 - f)^2 \alpha_x \alpha_y = (1 - f)(c + f \alpha_x \alpha_y) = (1 - f)\sqrt{v_x v_y}(r + f d_x d_y)$. Dividing the covariance by the square root of the product of the variances gives (2). (a) and (b) are read off, since the denominator is positive; with $P = d_x d_y > 0$, $r + fP = 0$ iff $\sqrt{P} = \sqrt{-r/f}$. For (c), divide numerator and denominator by $|d_x d_y|$: $r_{\text{coded}} = (r/|d_x d_y| + f \text{sign}(d_x d_y))/\sqrt{(d_x^{-2} + f)(d_y^{-2} + f)} \rightarrow \text{sign}(d_x d_y)$. For (d), the Cauchy-Schwarz inequality for the vectors $(1, \sqrt{f}|d_x|)$ and $(1, \sqrt{f}|d_y|)$ gives $(1 + f d_x^2)(1 + f d_y^2) \geq (1 + f|d_x||d_y|)^2$, with equality iff the vectors are equal, i.e. $|d_x| = |d_y|$. If $d_x d_y = P > 0$ the bound reads $|r_{\text{coded}}| \leq |G(\sqrt{P})|$, and r_{coded} and $G(\sqrt{P})$ have the same sign (that of $r + fP$). Hence $r_{\text{coded}} \leq G(\sqrt{P})$ if $r_{\text{coded}} > 0$ and $r_{\text{coded}} \geq G(\sqrt{P})$ if $r_{\text{coded}} < 0$. Since $G'(d) = 2fd(1 - r)/(1 + fd^2)^2 > 0$ for $d > 0$, G increases on $[0, \infty)$ from $G(0) = r$ to 1 , so $G(d_{\text{sym}}) = r_{\text{coded}}$ has a unique solution $d_{\text{sym}} \geq 0$ whenever $r \leq r_{\text{coded}} < 1$; solving $r + fd^2 = r_{\text{coded}}(1 + fd^2)$ gives the stated d_{sym}^2 , and the two inequalities become $d_{\text{sym}} \leq \sqrt{P}$

and $d_{\text{sym}} \geq \sqrt{P}$ respectively. (For $r_{\text{coded}} < 0$ the inequality $r_{\text{coded}} \geq G(\sqrt{P}) \geq G(0) = r$ shows that $r \leq r_{\text{coded}}$ holds automatically.) $\square$

Remark 3. The correlation itself does not depend on the divisor, but (2) is exact only when the present-row moments use divisor n_{11} and the pooled moments divisor n ; with divisors $n_{11} - 1$ and $n - 1$ it holds up to $O(1/n_{11})$. A coded or imputed table analysed as if complete also misstates precision: only the n_{11} jointly present rows carry information about the relation, and a standard error computed from n rows is too small [14, 15].

Proposition 4 (independent absences, mean imputation). *Let X be present on n_x rows, Y on n_y , both on n_{11} , and fill each absent value with its channel's present-data mean. Let $\bar{x}_{(x)}$, $V_{(x)}$ be the mean and variance of X over its n_x present rows (similarly $\bar{y}_{(y)}$, $V_{(y)}$), and let $\bar{x}_{11}$, $\bar{y}_{11}$, c_{11} and r_{cc} be the means, covariance and correlation over the jointly present rows. Then, exactly,*

$$r_{\text{imputed}} = \frac{n_{11} [c_{11} + (\bar{x}_{11} - \bar{x}_{(x)})(\bar{y}_{11} - \bar{y}_{(y)})]}{\sqrt{n_x V_{(x)} n_y V_{(y)}}}. \quad (3)$$

If the absences are missing completely at random [14] and the variables have finite fourth moments, then $r_{\text{imputed}} = r_{cc} n_{11} / \sqrt{n_x n_y} + O_p(n^{-1/2})$. If in addition the two absence indicators are independent with rates q_x, q_y , then $n_{11} / \sqrt{n_x n_y} \rightarrow \sqrt{(1 - q_x)(1 - q_y)}$ and $r_{\text{imputed}} \rightarrow \rho \sqrt{(1 - q_x)(1 - q_y)}$ almost surely, ρ the population correlation.

Proof. After imputation the overall mean of X is $(n_x \bar{x}_{(x)} + (n - n_x) \bar{x}_{(x)}) / n = \bar{x}_{(x)}$, so an imputed value has centred value zero and contributes nothing to any centred sum involving X ; the same holds for Y . The centred sum of squares of X is therefore $n_x V_{(x)}$ (that of Y is $n_y V_{(y)}$), and the centred cross-product reduces to the jointly present rows, $\sum_{11} (x_i - \bar{x}_{(x)})(y_i - \bar{y}_{(y)}) = n_{11} [c_{11} + (\bar{x}_{11} - \bar{x}_{(x)})(\bar{y}_{11} - \bar{y}_{(y)})]$, by writing $x_i - \bar{x}_{(x)} = (x_i - \bar{x}_{11}) + (\bar{x}_{11} - \bar{x}_{(x)})$ and similarly for y_i . Dividing by the square root of the product of the sums of squares gives (3). Under complete randomness each subset (jointly present rows, rows present in X , rows present in Y) is a random sample, so $\bar{x}_{11} - \bar{x}_{(x)}$ and $\bar{y}_{11} - \bar{y}_{(y)}$ are $O_p(n^{-1/2})$ and their product is $O_p(n^{-1})$, while $V_{(x)}$ and $V_{(y)}$ differ from the variances $V_{11,x}$, $V_{11,y}$ over the jointly present rows by $O_p(n^{-1/2})$ (finite fourth moments); since $r_{cc} = c_{11} / \sqrt{V_{11,x} V_{11,y}}$, this gives the second statement. With independent indicators, the strong law of large numbers gives $n_{11}/n \rightarrow (1 - q_x)(1 - q_y)$, $n_x/n \rightarrow 1 - q_x$, $n_y/n \rightarrow 1 - q_y$ and $r_{cc} \rightarrow \rho$ almost surely, which gives the third. $\square$

The identities (2) and (3) reproduce the directly computed correlations of simulated bivariate-normal data to rounding error.

Application: the air-quality record. The UCI Air Quality record of De Vito et al. [12, 13] (Section 7.1) has 9357 dated hourly rows. Each of its five metal-oxide sensor channels carries the code -200 in exactly 366 rows, and the union and the intersection of these absences are the same 366 rows: the five sensors are absent together. For $X = \text{PT08.S1(CO)}$ and $Y = \text{PT08.S3(NOx)}$ there are therefore $n_{11} = 8991$ present rows, 366 joint absences and no absence in one channel only, and $f = 366/9357 = 0.0391151$. Computed directly from the spreadsheet workbook of the data set, the present-row moments (divisor n_{11}) are

$$\mu_x = 1099.7079, \quad \sigma_x = 217.0725, \quad \mu_y = 835.3710, \quad \sigma_y = 256.8008, \quad r = -0.771918,$$

where $\sigma = \sqrt{v}$; with divisor $n_{11} - 1$ the standard deviations are 217.0846 and 256.8151. Both codes lie below both channel means (the readings are positive), so in the sign convention of Theorem 2 both distances are negative and $d_x d_y > 0$. Table 1 gives the distances and compares (2) with the Pearson correlation computed directly over all 9357 rows.

Table 1: The coded correlations of PT08.S1(CO) and PT08.S3(NOx), from the present-row moments by (2) and directly over all 9357 rows.

code $a_x = a_y$	d_x	d_y	$\sqrt{d_x d_y}$	closed form (2)	direct	difference
−200	−5.9874368	−4.0318055	4.9132658	+0.0869312	+0.0869312	9.7×10^{-17}
0	−5.0660856	−3.2529918	4.0595486	−0.0756296	−0.0756296	6.9×10^{-17}
channel means	0	0	0	−0.771918	−0.771918	0

The closed form reproduces both coded correlations, +0.086931 for the code −200 and −0.075630 for the code 0, to within 10^{-16} , and Theorem 2(a) accounts exactly for the equality of the mean-coded correlation with the present-row value −0.771918, because the absences are joint. The code −200 lies beyond the flip at $d^* = \sqrt{-r/f} = 4.4424$ and the code 0 short of it, so recoding the absences from −200 to 0 reverses the sign of the correlation. Theorem 2(d) turns each coded correlation into a bound on its code: $d_{\text{sym}} = 4.9038$ for $r_{\text{coded}} = +0.086931$ and 4.0681 for −0.075630, so, rounding outwards,

$$\sqrt{d_x d_y}|_{-200} \geq 4.903, \quad \sqrt{d_x d_y}|_0 \leq 4.069.$$

The directly computed distances satisfy both, $4.9133 \geq 4.9038$ and $4.0595 \leq 4.0681$, strictly, as the bound requires when $|d_x| \neq |d_y|$. The figures quoted belong to the workbook, which carries the hourly averages with fractional parts (mostly quarter units, such as 1292.25 and 1375.5). The semicolon-separated text version of the data set rounds the sensor channels to integers and gives $r = -0.771938, +0.087019$ (code −200) and −0.075529 (code 0).

Differences and correlations. The same decomposition explains why comparison against a reference removes a shared error while a shared code manufactures correlation. Let a sample and a standard be measured as $s_i + e_i$ and $t_i + e_i$ with a common, row-varying error e_i of variance σ_e^2 independent of s and t . The difference $(s_i + e_i) - (t_i + e_i)$ is free of e : the difference of a pair cancels its common part. The covariance, however, is $\text{cov}(s, t) + \sigma_e^2$: the common part adds to it and pulls the correlation towards +1. A shared code is the extreme case in which the common part is concentrated on the coded rows.

7 The sensor ladder

The same record carries a second pair, on the surfaces of its sensors. This section derives the two-state law of a metal-oxide sensor from mass action and reads it off the five sensors.

7.1 The record and its five surfaces

The UCI Air Quality data set [13], introduced by De Vito et al. [12], holds the hourly averaged responses of a gas multisensor device deployed at road level in a polluted area of an Italian city, together with hourly concentrations from a certified reference analyser. The data-set page describes 9358 instances recorded from March 2004 to February 2005; the spreadsheet workbook holds 9357 dated rows, from 10 March 2004 18:00 to 4 April 2005 14:00. Missing values are tagged −200. De Vito et al. calibrated the device for benzene with a neural network over a 13-month interval and reported seasonal effects at low concentration [12]. The data-set page identifies the channels as in Table 2.

NMHC(GT) is present only from 10 March to 1 May 2004 (one hour in May). There is no O₃ reference in the record, so S5 is read against the other gases.

Table 2: Channels of the air-quality record, as identified by the data-set page [13], and the number of hours (of 9357) on which each is present.

channel	material or instrument	nominal target, unit	hours present
PT08.S1	tin oxide (SnO ₂)	CO	8991
PT08.S2	titania (TiO ₂)	non-methane hydrocarbons (NMHC)	8991
PT08.S3	tungsten oxide (WO ₃)	NO _x	8991
PT08.S4	tungsten oxide (WO ₃)	NO ₂	8991
PT08.S5	indium oxide (In ₂ O ₃)	O ₃	8991
CO(GT)	reference analyser	CO, mg m ⁻³	7674
NMHC(GT)	reference analyser	NMHC, $\mu\text{g m}^{-3}$	914
C ₆ H ₆ (GT)	labelled reference analyser	benzene, $\mu\text{g m}^{-3}$	8991
NO _x (GT)	reference analyser	NO _x , ppb	7718
NO ₂ (GT)	reference analyser	NO ₂ , $\mu\text{g m}^{-3}$	7715
T, RH, AH	meteorology	°C, %, absolute humidity	8991

7.2 The two-state law of a metal-oxide surface

An n-type oxide grain carries adsorption sites, each in one of two states [9, 10]: “0”, free, with the electron in the conduction band; or “−”, holding an ionosorbed oxygen O[−] with one conduction electron trapped, which depletes the surface and raises its resistance. Let θ be the fraction of sites in state “−” and n_s the surface electron density. Three reactions exchange the two states, each at a mass-action rate that is first order in the fraction of sites in the reacting state:

- oxygen ionosorption $\frac{1}{2}\text{O}_2 + e^- + \text{S} \rightleftharpoons \text{O}_\text{S}^-$, forward rate $k_1 p_{\text{O}_2}^{1/2} n_s (1 - \theta)$, reverse rate $k_{-1} \theta$;
- release by a reducing gas R, $\text{R} + \text{O}_\text{S}^- \rightarrow \text{RO} + e^- + \text{S}$, rate $k_R [\text{R}] \theta$, which returns the electron;
- trapping by an oxidizing gas Ox, for example $\text{NO}_2 + e^- + \text{S} \rightarrow \text{NO} + \text{O}_\text{S}^-$, rate $k_O [\text{Ox}] n_s (1 - \theta)$, which traps an electron.

Proposition 5 (the surface pair). *Define the trapping coefficient $\Phi_- = (k_1 p_{\text{O}_2}^{1/2} + k_O [\text{Ox}]) n_s$ and the release coefficient $\Phi_+ = k_{-1} + k_R [\text{R}]$. At steady state*

$$\frac{\theta}{1 - \theta} = \frac{\Phi_-}{\Phi_+}, \quad \theta = \frac{1}{1 + e^\varepsilon}, \quad \varepsilon = \ln \frac{\Phi_+}{\Phi_-} = \ln \frac{1 - \theta}{\theta},$$

and every factor common to Φ_+ and Φ_- cancels from θ .

Proof. The fraction θ gains from ionosorption and trapping and loses to desorption and release, so $d\theta/dt = \Phi_- (1 - \theta) - \Phi_+ \theta$. Setting $d\theta/dt = 0$ gives $\Phi_- (1 - \theta) = \Phi_+ \theta$, hence $\theta/(1 - \theta) = \Phi_-/\Phi_+$ and $\theta = \Phi_-/(\Phi_- + \Phi_+) = 1/(1 + \Phi_+/\Phi_-) = 1/(1 + e^\varepsilon)$. If $\Phi_\pm = \lambda \phi_\pm$ with a common factor $\lambda > 0$, then $\Phi_+/\Phi_- = \phi_+/\phi_-$. $\square$

This is the pair (1) with $p = \theta$ and $\delta = \theta - \frac{1}{2} = -\frac{1}{2} \tanh(\varepsilon/2)$: which member holds the electron. It is exact for independent two-state sites under first-order mass action. The log-odds is linear in the log-ratio of the opposing gases under three conditions.

- (A1) Release by the reducing gas dominates thermal desorption: $k_R [\text{R}] \gg k_{-1}$.
- (A2) Trapping by the oxidizing gas dominates oxygen ionosorption: $k_O [\text{Ox}] \gg k_1 p_{\text{O}_2}^{1/2}$. If instead oxygen dominates, as in clean air where $p_{\text{O}_2} \approx 0.21$ bar is fixed, the oxidizing member of the pair is O₂ at fixed pressure and $\varepsilon = \ln[\text{R}] + \text{const}$.
- (A3) n_s does not depend on θ over the working range. In general $n_s = n_b e^{-qV_s/k_B T}$, where the depletion barrier V_s grows as $(N_s \theta)^2$ in the Schottky approximation, N_s being the site density; this feedback changes the effective exponent. Yamazoe and Shimano derive the

empirical power laws of semiconductor sensors from depletion theory: resistance $\propto P^n$, with $n \approx +1$ for NO_2 and O_3 and $n \approx -\frac{1}{2}$ for reducing gases such as H_2 and CO [11].

Under (A1)–(A3), $\Phi_+ \simeq k_R[\text{R}]$ and $\Phi_- \simeq k_O[\text{Ox}]n_s$. Allowing effective reaction orders β_R, β_O (1 for first order; other values for dissociative steps or the n_s feedback) and writing C_R, C_O for the two concentrations,

$$\varepsilon = \beta_R \ln C_R - \beta_O \ln C_O + \ln \frac{k_R}{k_O n_s}. \quad (4)$$

The reading. In the regime where the conductance changes linearly with the number of released electrons, the reading is $S = S_{\text{lo}} + (S_{\text{hi}} - S_{\text{lo}})(1 - \theta) = S_{\text{lo}} + (S_{\text{hi}} - S_{\text{lo}})/(1 + e^{-\varepsilon})$. With the reducing gas varying and the oxidizing gas fixed, (4) gives $e^{-\varepsilon} = (C_{1/2}/C)^\beta$ with $\beta = \beta_R$, and

$$S = S_{\text{lo}} + \frac{S_{\text{hi}} - S_{\text{lo}}}{1 + (C_{1/2}/C)^\beta}, \quad (5)$$

where the half-point $C_{1/2}$ is the concentration at which $\varepsilon = 0$: the release flux equals the trapping flux. With the oxidizing gas varying instead, $e^{-\varepsilon} = (C/C_{1/2})^{\beta_O}$, which is (5) with $\beta = \beta_O$ and $S_{\text{lo}}, S_{\text{hi}}$ exchanged. A sensor that falls with a reducing gas has $S_{\text{hi}} < S_{\text{lo}}$ in (5): its reading increases with θ , the opposite member of the same pair.

The power-law regime. For $C \ll C_{1/2}$, $S - S_{\text{lo}} \simeq (S_{\text{hi}} - S_{\text{lo}})(C/C_{1/2})^\beta \propto C^\beta$; for $C \gg C_{1/2}$ the reading saturates at S_{hi} . With $w = (C_{1/2}/C)^\beta$, $dS/d \ln C = (S_{\text{hi}} - S_{\text{lo}})\beta w/(1 + w)^2$, and since $S - S_{\text{lo}} = (S_{\text{hi}} - S_{\text{lo}})/(1 + w)$ and $S_{\text{hi}} - S = (S_{\text{hi}} - S_{\text{lo}})w/(1 + w)$, the log–log elasticity of the raw reading is

$$\frac{\partial \ln S}{\partial \ln C} = \beta \frac{(S - S_{\text{lo}})(S_{\text{hi}} - S)}{S(S_{\text{hi}} - S_{\text{lo}})}. \quad (6)$$

For a rising sensor with $0 < S_{\text{lo}} < S < S_{\text{hi}}$ both factors $(S - S_{\text{lo}})/S$ and $(S_{\text{hi}} - S)/(S_{\text{hi}} - S_{\text{lo}})$ lie in $(0, 1)$, so the elasticity is smaller than β : the baseline S_{lo} dilutes it.

Two gases: the balance line. Linearise $\ln S$ about an operating point. By (4), $\partial \ln S / \partial \ln C_R = \psi \beta_R$ and $\partial \ln S / \partial \ln C_O = -\psi \beta_O$ with the same $\psi = d \ln S / d \varepsilon$. When the regressors of $\ln S$ are the reducing and the oxidizing member themselves, the pair law therefore gives coefficients b_R, b_O of opposite sign, $\text{sign } b_R = -\text{sign } b_O$, and a constant reading along the balance line $d \ln C_O / d \ln C_R = -b_R / b_O = \beta_R / \beta_O > 0$, of slope 1 for equal orders. The reading depends on the gases only through ε , and under (A1)–(A3) ε is the log-ratio, the odd part, of the two opposing fluxes.

Humidity. A term that multiplies Φ_+ and Φ_- equally is common-mode and cancels in ε by Theorem 5. A water term that donates electrons adds to Φ_+ and is not common-mode. Bârsan and Weimar analyse the effect of water vapour on CO sensing by SnO_2 [24]. The record decides between the two cases in Section 7.8.

7.3 Statistical methods

- *Rows.* For a sensor–gas pair, all hours on which both are present (pairwise-complete).
- *Correlations.* Pearson r and Spearman ρ .
- *Elasticity.* The ordinary least-squares (OLS) slope of $\ln S$ on $\ln C$.
- *Two-state fit.* Nonlinear least squares of S on C for (5) (four parameters $S_{\text{lo}}, S_{\text{hi}}, C_{1/2}, \beta$), minimising the residual sum of squares from multiple starting values, with $\beta \in [0.05, 8]$ and $\ln C_{1/2}$ within 6 of the observed range of $\ln C$.

- *Comparison model.* The $C_{1/2} \rightarrow \infty$ limit of (5) at fixed $(S_{\text{hi}} - S_{\text{lo}})C_{1/2}^{-\beta}$, the offset power law $S = S_{\text{lo}} + \mathcal{A}C^\beta$ (three parameters), compared through $\text{AIC} = n \ln(\text{RSS}/n) + 2k$ with k the number of parameters; ΔAIC is that of the power law minus that of the two-state law.
- *Uncertainty.* Hourly data are serially correlated, so all intervals come from a day-block bootstrap [25] with the calendar day as block: days are resampled with replacement and every statistic is recomputed. There are 1000 replicates unless stated; the seasonal half-point fits use 400 and the S4 CO/NO ratio 2000. Intervals are the 2.5 and 97.5 percentiles. Naive OLS intervals are also given for the elasticities.
- *Identification of $C_{1/2}$.* A half-point is identified when its 95% bootstrap interval lies inside the observed concentration range.
- *Pair regression.* $\ln S_k = b_0 + b_R \ln \text{CO} + b_O \ln \text{NO}_2 + b_H \ln \text{AH}$ for sensor k , fitted by OLS on the 6941 hours (341 days) on which CO, NO₂, AH and all sensors are present.
- *NO.* $\text{NO} = \text{NO}_x - \text{NO}_2/1.9125$ in ppb, since 1 ppb of NO₂ is $1.9125 \mu\text{g m}^{-3}$ at 20 °C and 1013.25 hPa (molar mass $46.005 \text{ g mol}^{-1}$ from the standard atomic weights [26], molar volume RT/P [22]); the 1 hour with $\text{NO} \leq 0$ is excluded. The conventional 25 °C factor 1.8816 (molar volume 24.45 L mol^{-1}) changes every coefficient by less than 0.002.
- *Seasons.* Warm half April–September, cold half October–March.

7.4 Correlations: four rising members and one falling

The number of pairwise-present hours depends on the gas: CO 7344, NMHC 887, C₆H₆ 8991, NO_x 7396, NO₂ 7393, and T, RH, AH 8991. Tables 3 and 4 give the correlations.

Table 3: Pearson correlation of each sensor with each reference and meteorological channel.

sensor	CO	NMHC	C ₆ H ₆	NO _x	NO ₂	T	RH	AH
S1 SnO ₂	+0.879	+0.791	+0.884	+0.714	+0.642	+0.049	+0.115	+0.135
S2 TiO ₂	+0.916	+0.878	+0.982	+0.704	+0.647	+0.241	−0.090	+0.187
S3 WO ₃	−0.703	−0.771	−0.736	−0.656	−0.652	−0.145	−0.057	−0.232
S4 WO ₃	+0.631	+0.853	+0.766	+0.234	+0.158	+0.561	−0.032	+0.630
S5 In ₂ O ₃	+0.854	+0.767	+0.866	+0.787	+0.708	−0.027	+0.125	+0.071

Table 4: Spearman correlation of each sensor with each reference and meteorological channel.

sensor	CO	NMHC	C ₆ H ₆	NO _x	NO ₂	T	RH	AH
S1	+0.879	+0.850	+0.890	+0.725	+0.674	+0.081	+0.092	+0.138
S2	+0.927	+0.950	+1.000	+0.705	+0.670	+0.276	−0.122	+0.192
S3	−0.811	−0.950	−0.850	−0.790	−0.689	−0.118	−0.084	−0.217
S4	+0.594	+0.895	+0.749	+0.160	+0.154	+0.615	−0.066	+0.645
S5	+0.856	+0.804	+0.874	+0.793	+0.709	−0.002	+0.128	+0.083

Sign pattern. S1, S2, S4 and S5 rise with every traffic gas (CO, NMHC, NO_x, NO₂); the tungsten-oxide sensor S3 falls with every one. The same holds for absolute humidity: S1, S2, S4 and S5 rise, S3 falls. In the two-state law S3 is the same pair read from the opposite member: it has $S_{\text{hi}} < S_{\text{lo}}$ in (5), and its occupancy $(S - S_{\text{lo}})/(S_{\text{hi}} - S_{\text{lo}})$ rises with the gas as the others' does. S4 is the sensor most tied to the weather, with $r = +0.561$ against T and +0.630 against AH. The Spearman coefficient +1.000 of S2 with C₆H₆ is exact, and Section 7.9 explains it.

The reference gases are strongly collinear (Table 5), which limits what any regression on them can separate.

Table 5: Correlations of the logarithms of the reference channels.

pair	correlation	hours	pair	correlation	hours
CO–NMHC	0.921	890	NMHC–NO ₂	0.823	877
CO–NO _x	0.773	7261	NO _x –NO ₂	0.851	7715
CO–NO ₂	0.719	7258	CO–AH	0.078	7344
NMHC–NO _x	0.877	877	NO ₂ –AH	−0.289	7393

7.5 Elasticities

Table 6 gives the log–log elasticities $\partial \ln S / \partial \ln C$ of every sensor to every gas and to absolute humidity. Against its nominal gas, S1–CO is +0.221 [0.209, 0.233] ($R^2 = 0.717$), S2–NMHC +0.281 [0.268, 0.294] ($R^2 = 0.890$), S3–NO_x −0.267 [−0.282, −0.251] ($R^2 = 0.615$) and S4–NO₂ +0.087 [0.059, 0.114] ($R^2 = 0.027$). No O₃ is measured; the strongest reference of S5 is CO, +0.458 [0.433, 0.484] ($R^2 = 0.671$). The day-block intervals are 1.86 to 4.20 times wider than the naive ones (narrowest ratio S3–NMHC, widest S2–CO), which is the effect of serial correlation.

Table 6: Log–log elasticities of the five sensors, with naive and day-block bootstrap 95% intervals.

sensor	gas	elasticity	naive 95%	day-block 95%	R^2	hours	days
S1	CO	+0.2210	[+0.2178, +0.2241]	[+0.2086, +0.2327]	0.7174	7344	347
S1	NMHC	+0.1708	[+0.1635, +0.1781]	[+0.1549, +0.1845]	0.7025	887	44
S1	NO _x	+0.1589	[+0.1555, +0.1622]	[+0.1512, +0.1664]	0.5370	7396	349
S1	NO ₂	+0.2644	[+0.2577, +0.2711]	[+0.2458, +0.2843]	0.4459	7393	349
S1	AH	+0.0785	[+0.0698, +0.0872]	[+0.0541, +0.1024]	0.0335	8991	383
S2	CO	+0.3451	[+0.3412, +0.3491]	[+0.3283, +0.3614]	0.7999	7344	347
S2	NMHC	+0.2814	[+0.2749, +0.2880]	[+0.2680, +0.2935]	0.8898	887	44
S2	NO _x	+0.2324	[+0.2273, +0.2375]	[+0.2214, +0.2439]	0.5217	7396	349
S2	NO ₂	+0.3979	[+0.3880, +0.4077]	[+0.3636, +0.4287]	0.4583	7393	349
S2	AH	+0.1469	[+0.1340, +0.1598]	[+0.1119, +0.1794]	0.0524	8991	383
S3	CO	−0.3201	[−0.3259, −0.3143]	[−0.3403, −0.2999]	0.6134	7344	347
S3	NMHC	−0.2646	[−0.2705, −0.2587]	[−0.2753, −0.2534]	0.8976	887	44
S3	NO _x	−0.2670	[−0.2718, −0.2622]	[−0.2823, −0.2511]	0.6153	7396	349
S3	NO ₂	−0.4223	[−0.4327, −0.4119]	[−0.4587, −0.3856]	0.4615	7393	349
S3	AH	−0.1467	[−0.1602, −0.1332]	[−0.1840, −0.1065]	0.0480	8991	383
S4	CO	+0.1961	[+0.1896, +0.2026]	[+0.1789, +0.2150]	0.3212	7344	347
S4	NMHC	+0.1654	[+0.1595, +0.1712]	[+0.1539, +0.1771]	0.7745	887	44
S4	NO _x	+0.0531	[+0.0466, +0.0596]	[+0.0391, +0.0679]	0.0334	7396	349
S4	NO ₂	+0.0865	[+0.0745, +0.0984]	[+0.0591, +0.1140]	0.0266	7393	349
S4	AH	+0.4038	[+0.3958, +0.4118]	[+0.3795, +0.4260]	0.5206	8991	383
S5	CO	+0.4577	[+0.4504, +0.4651]	[+0.4329, +0.4839]	0.6710	7344	347
S5	NMHC	+0.3431	[+0.3256, +0.3605]	[+0.3039, +0.3818]	0.6263	887	44
S5	NO _x	+0.3657	[+0.3591, +0.3722]	[+0.3519, +0.3801]	0.6175	7396	349
S5	NO ₂	+0.6021	[+0.5884, +0.6158]	[+0.5571, +0.6501]	0.5017	7393	349
S5	AH	+0.1270	[+0.1083, +0.1457]	[+0.0606, +0.1937]	0.0194	8991	383

7.6 Two-state fits against the nominal gas

Table 7 fits (5) to each sensor against its nominal gas, and S5 against CO. The concentration quantiles (minimum, 5%, 50%, 95%, maximum) are: CO 0.1, 0.5, 1.8, 4.9, 11.9 mg m^{−3}; NMHC 7, 28, 145, 667.5, 1189 μ g m^{−3}; NO_x 2, 37, 178, 677.25, 1479 ppb; NO₂ 2, 43, 108.5, 198, 332.6 μ g m^{−3}.

The half-point of S5 against CO is identified. That of S1 against CO lies at the upper edge of the data: 95.2% of its replicates fall inside the observed range of CO and 0.3% inside its central 90%. The half-points of S2 against NMHC and of S3 against NO_x are not identified (for

Table 7: Two-state fits (5) against the nominal gas, with day-block bootstrap 95% intervals, and ΔAIC against the offset power law.

pair (unit of C)	hours	S_{lo}	S_{hi}	$C_{1/2}$ [95%]	β [95%]	RMSE	ΔAIC
S1-CO (mg m^{-3})	7344	793.7	2607.0	7.494 [5.41, 13.5]	1.184 [1.005, 1.357]	100.9	82.7
S2-NMHC ($\mu\text{g m}^{-3}$)	887	308.9	2281.6	495.7 [248, 1.1×10^5]	0.648 [0.300, 0.929]	92.4	6.3
S3-NO _x (ppb)	7396	2100.3	-351.0	136.0 [1.81, 6.6×10^4]	0.364 [0.100, 0.619]	170.9	8.2
S4-NO ₂ ($\mu\text{g m}^{-3}$)	7393	1256.8	1519.1	67.98 [48.8, 88.2]	3.85 [1.83, 8.00]	346.6	36.3
S5-CO (mg m^{-3})	7344	511.2	2802.1	4.611 [3.90, 5.67]	1.445 [1.290, 1.615]	202.9	162.5

S3 the fitted S_{hi} is negative). S4 against NO₂ is located, with the upper end of its β interval at the bound 8, but its step (262) is smaller than its RMSE and its R^2 is 0.027. The offset power-law exponents are 0.720 for S1-CO ($S_{\text{lo}} = 713.0$), 0.238 for S2-NMHC and 0.688 for S5-CO; for S3-NO_x and S4-NO₂ the power law degenerates ($|\beta| < 0.04$). The bootstrap draws of Table 7 and Table 8 are independent, which is why the intervals of S1 and S5 against CO differ between the two tables in the third digit.

7.7 The ladder on the common CO axis

Fitting all five sensors against the same gas, CO, on its 7344 hours (347 days), places them on one axis (Tables 8 and 9, Fig. 2d).

Table 8: Two-state fits of all five sensors against CO, with day-block bootstrap 95% intervals.

sensor	direction	S_{lo}	S_{hi}	$C_{1/2}$ (mg m^{-3}) [95%]	β [95%]	RMSE	ΔAIC
S3	falls	1340.5	322.0	1.682 [1.392, 1.976]	1.242 [1.005, 1.484]	165.0	147.4
S5	rises	511.2	2802.1	4.611 [3.909, 5.716]	1.445 [1.292, 1.611]	202.9	162.5
S2	rises	519.8	2394.3	5.399 [4.194, 7.493]	1.211 [1.058, 1.358]	97.6	209.0
S1	rises	793.7	2607.0	7.494 [5.341, 13.51]	1.184 [1.013, 1.372]	100.9	82.7
S4	rises	1105.3	3277.5	7.994 [4.406, 37.38]	1.242 [0.918, 1.628]	270.8	11.7

Table 9: The comparison fits on the CO axis, and the share of bootstrap replicates of $C_{1/2}$ inside the full range of hourly CO and inside its central 90%.

sensor	offset power law ($S_{\text{lo}}, \mathcal{A}$, exponent)	inside [0.1, 11.9]	inside [0.5, 4.9]
S3	(2299.5, -1324.4, 0.187)	1.000	1.000
S5	(293.4, 467.0, 0.688)	1.000	0.740
S2	(372.6, 375.2, 0.631)	1.000	0.229
S1	(713.0, 240.8, 0.720)	0.946	0.006
S4	(1025.6, 240.2, 0.784)	0.769	0.078

The two-state law is preferred over the offset power law for every sensor, with ΔAIC from 11.7 (S4) to 209.0 (S2). The fitted slopes $\beta = 1.18$ to 1.45 lie just above first-order mass action: the intervals exclude 1 for S1, S2, S3 and S5 and include it for S4. The half-points order the sensors into a ladder,

$$\text{S3 (1.68, falling)} < \text{S5 (4.61)} < \text{S2 (5.40)} < \text{S1 (7.49)} < \text{S4 (7.99)} \quad \text{mg m}^{-3} \text{ of CO.}$$

The half-points of S3, S5 and S2 are identified inside the observed range; that of S1 has its interval reaching above the maximum hourly CO, and that of S4 is not identified. Ordered by elasticity to CO instead, the sensors read $\text{S3 } -0.320 < \text{S4 } +0.196 < \text{S1 } +0.221 < \text{S2 } +0.345 < \text{S5 } +0.458$; the elasticity mixes the half-point with the baseline dilution of (6), and the half-point separates them.

Stability across the year. Table 10 repeats the fits in the two halves of the year; the maximum hourly CO is 7.5 mg m^{-3} in the warm half and 11.9 in the cold half. The rung order $S3 < S5 < S2$ holds in both halves. S1 is identified in the cold half only. The elasticities to the nominal gas in the warm and cold halves are: S1–CO 0.213 [0.200, 0.226] and 0.222 [0.203, 0.241]; S5–CO 0.433 [0.405, 0.459] and 0.467 [0.425, 0.508]; S5–NO₂ 0.557 [0.492, 0.624] and 0.672 [0.616, 0.727]; S3–NO_x -0.237 [-0.259 , -0.217] and -0.370 [-0.384 , -0.354]; S4–NO₂ 0.194 [0.167, 0.221] and 0.153 [0.111, 0.195]. S2–NMHC is 0.300 in April and 0.277 in March, the only months with NMHC.

Table 10: Half-points on the CO axis in the warm (April–September) and cold (October–March) halves of the year, with day-block bootstrap 95% intervals (400 replicates).

sensor	warm $C_{1/2}$ [95%]	warm β	cold $C_{1/2}$ [95%]	cold β
S3	1.442 [0.958, 1.979]	0.919	1.654 [1.262, 2.007]	1.512
S5	3.871 [3.109, 5.288]	1.648	4.599 [3.818, 6.143]	1.378
S2	4.781 [3.746, 8.070]	1.186	5.668 [4.314, 8.067]	1.273
S1	18.89 [5.84, 74.1] (not identified)	0.958	6.132 [4.709, 9.823]	1.293
S4	7.757 [4.29, 32.4] (not identified)	1.274	15.36 [6.18, 42.3] (not identified)	1.090

7.8 Two gases and humidity

The pair regression of Section 7.3 reads the balance line of Section 7.2 with CO as the reducing and NO₂ as the candidate oxidizing marker, over 6941 hours on 341 days. The regressors are collinear: the correlation of $\ln \text{CO}$ and $\ln \text{NO}_2$ is 0.713, of $\ln \text{CO}$ and $\ln \text{AH}$ 0.092, and of $\ln \text{NO}_2$ and $\ln \text{AH}$ -0.308 ; the variance inflation factors are 2.60 ($\ln \text{CO}$), 2.85 ($\ln \text{NO}_2$) and 1.41 ($\ln \text{AH}$). Table 11 gives the coefficients, each with its 95% interval on the line below.

Table 11: Pair regression $\ln S = b_0 + b_R \ln \text{CO} + b_O \ln \text{NO}_2 (+ b_H \ln \text{AH})$, 6941 hours, with the humidity term and without it (“no AH”). Second line of each entry: day-block bootstrap 95% interval. The last-but-one column gives whether b_R and b_O have opposite signs, with the share of replicates in which they do.

sensor	model	b_R (CO)	b_O (NO ₂)	b_H (AH)	$-b_R/b_O$	opposite signs	R^2
S1	with AH	+0.1610 [+0.1360, +0.1889]	+0.1194 [+0.0789, +0.1575]	+0.1019 [+0.0791, +0.1213]	-1.348 [-2.422 , -0.875]	no (0.000)	0.7573
S2	with AH	+0.2662 [+0.2356, +0.3025]	+0.1499 [+0.1000, +0.2005]	+0.1614 [+0.1381, +0.1859]	-1.776 [-3.017 , -1.186]	no (0.000)	0.8456
S3	with AH	-0.1692 [-0.2103 , -0.1321]	-0.2990 [-0.3678 , -0.2308]	-0.2130 [-0.2477 , -0.1807]	-0.566 [-0.895 , -0.365]	no (0.000)	0.7030
S4	with AH	+0.1835 [+0.1616, +0.2113]	-0.0073 [-0.0462 , +0.0218]	+0.3829 [+0.3570, +0.4054]	+25.0 [-274.5 , +101.8]	b_O null (0.672)	0.7932
S5	with AH	+0.2839 [+0.2355, +0.3305]	+0.3570 [+0.2783, +0.4423]	+0.2071 [+0.1588, +0.2515]	-0.795 [-1.173 , -0.533]	no (0.000)	0.7361
S1	no AH	+0.2016 [+0.1794, +0.2275]	+0.0448 [+0.0116, +0.0748]	–	-4.50 [-17.6 , -2.35]	no (0.004)	0.7156
S2	no AH	+0.3305 [+0.2970, +0.3682]	+0.0317 [-0.0123 , +0.0737]	–	-10.4 [-174.6 , +105.4]	b_O null (0.092)	0.7969
S3	no AH	-0.2540 [-0.2950 , -0.2210]	-0.1430 [-0.1965 , -0.0889]	–	-1.777 [-3.353 , -1.117]	no (0.000)	0.6283
S4	no AH	+0.3360 [+0.2944, +0.3855]	-0.2878 [-0.3650 , -0.2201]	–	+ 1.167 [+1.040, +1.365]	yes (1.000)	0.4650
S5	no AH	+0.3664 [+0.3251, +0.4133]	+0.2054 [+0.1423, +0.2666]	–	-1.784 [-2.904 , -1.239]	no (0.000)	0.6983

What the regression gives. In this urban record NO₂ rises with CO. Both are emitted by traffic (the correlation of their logarithms is 0.713 over these hours), and NO₂ also follows

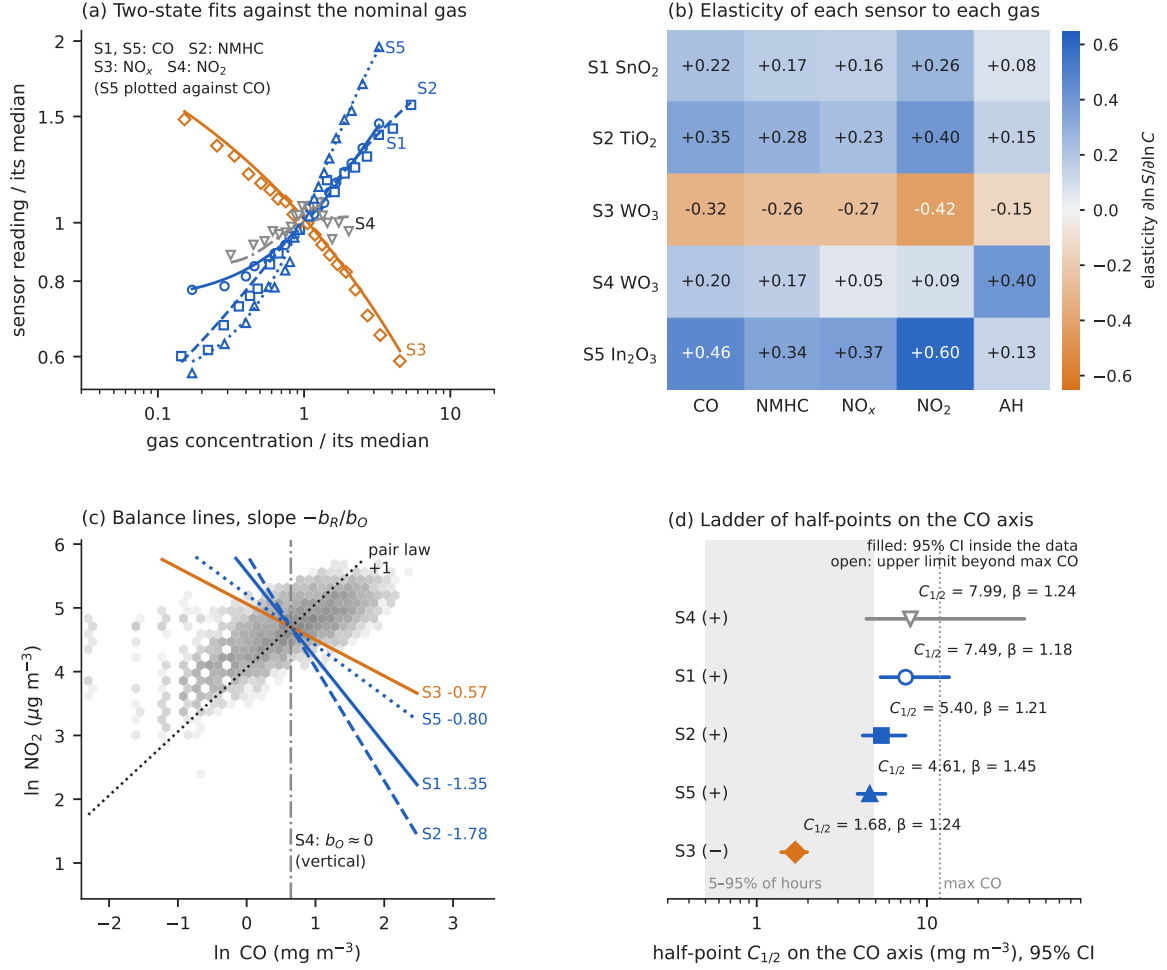


Figure 2: The sensor ladder of the UCI air-quality record (8991 hours with all five metal-oxide sensors present). Blue marks the sensors whose reading rises with the traffic gases (S1 tin oxide, S2 titania, S5 indium oxide), orange the tungsten-oxide sensor S3, which falls, and gray the tungsten-oxide sensor S4. (a) Binned medians (20 equal-count bins, open markers) of each sensor against its nominal reference gas, both normalised by their medians, with the fitted two-state law $S = S_{lo} + (S_{hi} - S_{lo})/[1 + (C_{1/2}/C)^\beta]$ (lines). Pairs: S1–CO, S2–NMHC (887 hours, March–April 2004), S3–NO_x, S4–NO₂, S5–CO (the record has no O₃ reference). (b) Log–log elasticities $\partial \ln S / \partial \ln C$ of the five sensors to CO, NMHC, NO_x, NO₂ and absolute humidity, from least-squares slopes over pairwise-present hours. (c) Balance lines in the $(\ln \text{CO}, \ln \text{NO}_2)$ plane, along which the fitted pair regression $\ln S = b_0 + b_R \ln \text{CO} + b_O \ln \text{NO}_2 + b_H \ln \text{AH}$ gives a constant reading (slope $-b_R/b_O$, drawn through the median point). The dotted line is the slope +1 of the pair law with equal orders. The S4 line is vertical because its b_O is null. Gray hexagons show the density of the 6941 jointly present hours. (d) The ladder: half-points $C_{1/2}$ of the two-state fits on the common CO axis, with day-block bootstrap 95% intervals, ordered S3 (1.68 mg m⁻³) < S5 (4.61) < S2 (5.40) < S1 (7.49) < S4 (7.99). Filled markers: interval inside the observed CO range. Open markers: upper end beyond the maximum hourly CO (11.9 mg m⁻³, dotted). Shading: central 90% of hourly CO. Labels give β .

the other reducing gases of the traffic plume (NMHC–NO₂ 0.823, Table 5), so the CO and NO₂ coefficients carry the same sign on S1, S2, S3 and S5, in every bootstrap replicate, and their balance lines have negative slopes (−1.348, −1.776, −0.566 and −0.795). The oxidizing member of the surface pair is therefore not separable through NO₂ in this record. On S4 the NO₂ coefficient is null once humidity enters (−0.0073 [−0.0462, +0.0218]). Without the humidity term S4 has $b_R = +0.336$ and $b_O = -0.288$, of opposite sign in every replicate, and balance slope +1.167 [1.040, 1.365], close to the slope 1 of equal orders; the humidity term absorbs this structure because $\ln \text{NO}_2$ and $\ln \text{AH}$ are correlated (−0.308) and S4 is dominated by humidity (elasticity +0.404, $R^2 = 0.521$). The same pattern holds in both halves of the year: only the warm-half b_O of S1 touches zero (+0.032 [−0.005, +0.059]), and the b_O of S4 is null in both halves (warm, 3056 hours: −0.003 [−0.071, +0.040]; cold, 3885 hours: +0.036 [−0.021, +0.084]).

The opposite-sign pair: CO against NO on S4. Separating NO from NO₂ gives the regression $\ln S = b_0 + b_1 \ln \text{CO} + b_2 \ln \text{NO} + b_3 \ln \text{NO}_2 + b_4 \ln \text{AH}$ over 6940 hours (Table 12); the regressor correlations are 0.738 (CO–NO), 0.713 (CO–NO₂) and 0.790 (NO–NO₂).

Table 12: Regression with NO and NO₂ separated, 6940 hours. Second line: day-block bootstrap 95% interval.

sensor	CO	NO	NO ₂	AH	R^2
S1	+0.1441 [+0.1187, +0.1728]	+0.0256 [+0.0141, +0.0369]	+0.0896 [+0.0622, +0.1166]	+0.1068 [+0.0867, +0.1272]	0.7655
S2	+0.2575 [+0.2227, +0.3001]	+0.0129 [−0.0012, +0.0251]	+0.1356 [+0.0935, +0.1810]	+0.1640 [+0.1396, +0.1875]	0.8467
S3	−0.0986 [−0.1285, −0.0710]	−0.1082 [−0.1263, −0.0884]	−0.1704 [−0.2213, −0.1297]	−0.2331 [−0.2612, −0.2055]	0.7597
S4	+0.2286 [+0.2002, +0.2680]	−0.0698 [−0.0822, −0.0584]	+0.0772 [+0.0415, +0.1094]	+0.3702 [+0.3468, +0.3929]	0.8248
S5	+0.2122 [+0.1741, +0.2539]	+0.1099 [+0.0907, +0.1288]	+0.2263 [+0.1691, +0.2899]	+0.2274 [+0.1840, +0.2700]	0.7672

The opposite-sign pair of the record appears on the tungsten-oxide sensor S4, as CO (+0.229) against NO (−0.070): its balance slope in the ($\ln \text{CO}, \ln \text{NO}$) plane is 3.28 [2.90, 3.73], with opposite signs in all 2000 replicates. S3 falls with CO also when NO and NO₂ are held fixed (−0.099 [−0.129, −0.071]): its opposite direction is its own response to the reducing gas, not an effect of the co-emitted nitrogen oxides, and all its gas coefficients share one sign.

Humidity. In Table 11 the humidity coefficient b_H has the sign of b_R in all five sensors (S1 +0.102, S2 +0.161, S3 −0.213, S4 +0.383, S5 +0.207), and every interval excludes zero. The interaction model $\ln S = b_0 + b_1 \ln \text{CO} + b_2 \ln \text{NO}_2 + b_3 \omega + b_4 \omega \ln \text{CO} + b_5 \omega \ln \text{NO}_2$, with $\omega = \ln \text{AH} - \bar{\ln \text{AH}}$, gives Table 13.

Humidity is therefore not a common-mode term in this record. A common-mode term would cancel in ε (Theorem 5), leave the gas elasticities unchanged and have no interactions. Instead absolute humidity shifts every reading in the direction of the reducing gas, which is the mass-action signature of an electron-donating co-adsorbate on the release side Φ_+ ; it modulates the CO elasticity (interaction intervals exclude zero for S1, S2, S3 and S4) and the NO₂ elasticity (intervals exclude zero for S2, S3, S4 and S5).

Table 13: Humidity interaction model. Second line: day-block bootstrap 95% interval.

sensor	b_1	b_2	b_3	b_4 (CO $\times$ AH)	b_5 (NO $_2\times$ AH)	R^2
S1	+0.1646 [+0.1420, +0.1915]	+0.1152 [+0.0781, +0.1518]	+0.2020 [+0.0323, +0.3574]	+0.0535 [+0.0255, +0.0771]	−0.0270 [−0.0622, +0.0112]	0.7625
S2	+0.2714 [+0.2380, +0.3051]	+0.1452 [+0.0992, +0.1926]	+0.6432 [+0.3736, +0.8841]	+0.0648 [+0.0087, +0.1079]	−0.1101 [−0.1663, −0.0473]	0.8497
S3	−0.1838 [−0.2219, −0.1486]	−0.2868 [−0.3516, −0.2209]	−1.7370 [−2.0429, −1.4709]	−0.1742 [−0.2169, −0.1322]	+0.3452 [+0.2867, +0.4119]	0.7362
S4	+0.1797 [+0.1584, +0.2079]	−0.0037 [−0.0448, +0.0260]	+0.0782 [−0.1934, +0.3135]	−0.0494 [−0.0996, −0.0079]	+0.0705 [+0.0187, +0.1331]	0.7955
S5	+0.2898 [+0.2485, +0.3328]	+0.3546 [+0.2799, +0.4288]	+1.3845 [+0.9741, +1.7971]	+0.0478 [−0.0241, +0.1158]	−0.2578 [−0.3514, −0.1605]	0.7494

7.9 The benzene column is a transform of S2

Over all 8991 present rows the Spearman correlation of S2 with C₆H₆(GT) is exactly 1: sorted by S2, the benzene column never decreases. Least squares of $\ln C_6H_6$ on $\ln(S_2 - b)$ gives

$$C_6H_6 = 1.23978 \times 10^{-4} (S_2 - 324.785)^{1.743158} \quad (\mu\text{g m}^{-3}), \quad (7)$$

with maximum relative deviation 1.6×10^{-15} and root-mean-square relative deviation 5.0×10^{-16} at the least-squares values of the three parameters, over $S_2 \in [383.25, 2214.0]$ and $C_6H_6 \in [0.149, 63.74]$; with the parameters rounded as printed, (7) reproduces every value to a relative deviation of 4.5×10^{-6} . The column that the data-set page describes as the true hourly benzene concentration from the reference analyser is therefore a deterministic calibration of the titania sensor, not an independent measurement. It carries no information beyond S2: a model that predicts C₆H₆(GT) from the sensor channels recovers a known function of one of its own inputs, and because the transform is strictly increasing, every rank statistic of C₆H₆(GT) equals that of S2. We exclude it as a reference from all fits. Its form is exactly the power-law regime of (5), $S_2 - S_{lo} \propto C^\beta$ with $S_{lo} = 324.785$ and $\beta = 1/1.743158 = 0.5737$; for comparison, the offset power law of S2 against CO has exponent 0.631 with $S_{lo} = 372.6$ (Table 9).

7.10 Summary of the sensor results

Every sensor reading is a monotone, saturating function of the traffic-gas level, and the two-state logistic in $\ln C$ fits it better than the offset power law, with slopes $\beta = 1.18$ to 1.45 near first-order mass action. On the common CO axis the half-points form the ladder S3 (1.68) < S5 (4.61) < S2 (5.40) mg m^{−3}, identified inside the data and ordered in the same way in both halves of the year; the half-point of S1 (7.49) lies at the upper edge of the data, and that of S4 is not identified. S3 reads the opposite member of the pair, and it falls with CO also when NO and NO₂ are held fixed. Humidity is a release-side term, not a common-mode one. In this urban record NO₂ rises with CO, so the CO and NO₂ coefficients carry the same sign on S1, S2, S3 and S5 and the oxidizing member is not separable through NO₂; the opposite-sign pair is CO against NO on S4, with balance slope 3.28 [2.90, 3.73]. The benzene column is the provider’s calibration of S2, in the power-law regime of the two-state law. The record has no O₃ reference, so S5 is read against CO.

7.11 The same law in the giant planets

The same two-state law, reached through a mass-action balance between two opposing species as in Theorem 5, decides the top ice deck of the giant planets [18]. There NH₃ and H₂S are consumed 1:1 by the NH₄SH lattice, and at every level where the solid coexists with the gas their mole fractions satisfy $x_N x_S = K(T)/P^2$, with $K(T)$ the NH₄SH dissociation constant in

atm² and P the pressure in atm. Write $\chi = \sqrt{K}/P$ for the common part and $x_N = \chi e^{\varepsilon/2}$, $x_S = \chi e^{-\varepsilon/2}$; then $x_N x_S = \chi^2$ holds identically, and the difference

$$\Delta = x_N - x_S = 2\chi \sinh(\varepsilon/2), \quad \varepsilon = 2 \operatorname{arsinh} \frac{\Delta}{2\chi} = \ln \frac{x_N}{x_S},$$

is conserved from the deep atmosphere upwards because the solid takes the two gases 1:1. The NH₃ share of the vapour is $\phi_N = x_N/(x_N + x_S) = 1/(1 + e^{-\varepsilon})$, with $\phi_N - \frac{1}{2} = \frac{1}{2} \tanh(\varepsilon/2)$. At the deck base the deep mole fractions N and S are still intact, so $\chi = \sqrt{NS}$ and $\varepsilon = \ln(N/S)$; above it K falls on cooling, $\chi \rightarrow 0$ and $\varepsilon \rightarrow \operatorname{sign}(\Delta) \cdot \infty$. The surviving member is $\operatorname{sign}(N - S)$, with mole fraction $|\Delta|$, and at $N = S$ neither survives. The deep N/S ratios are 4.73 (Jupiter), 3.16 (Saturn), 0.195 (Uranus) and 0.170 (Neptune) [18]: the zero $N/S = 1$ separates the gas giants, above whose NH₄SH deck NH₃ survives, from the ice giants, above whose deck H₂S survives. Table 14 sets the two systems side by side.

Table 14: One pair law on a sensor surface and at the NH₄SH deck of a giant planet.

	metal-oxide sensor (Section 7.2)	NH ₄ SH deck [18]
pair	a site holds the electron (O ⁻) or releases it	the surviving vapour is NH ₃ or H ₂ S
balance	$\theta/(1 - \theta) = \Phi_-/\Phi_+$	$x_N/x_S = e^\varepsilon$ with $x_N x_S = K/P^2$
log-odds	$\varepsilon = \ln(\Phi_+/\Phi_-) = \beta_R \ln C_R - \beta_O \ln C_O + \text{const}$	$\varepsilon = 2 \operatorname{arsinh}(\Delta P/(2\sqrt{K}))$, equal to $\ln(N/S)$ at the base
occupancy	$1 - \theta = 1/(1 + e^{-\varepsilon})$	$\phi_N = 1/(1 + e^{-\varepsilon})$
common part (cancels)	any factor common to Φ_+ and Φ_-	$\chi = \sqrt{K}/P$, the geometric mean
zero	$C = C_{1/2}$: release flux equals trapping flux	$N = S$: neither gas survives
what the observer reads	which member holds the electron	which gas survives the deck

Both are instances of one mechanism: two opposing reactants compete for a shared partner, and mass action converts their ratio into an occupancy. On the sensor, reducing and oxidizing gases compete for the surface oxygen and its trapped electron; in the giants, NH₃ and H₂S are consumed together by the NH₄SH lattice. The sensor is an open steady state whose log-ratio is set by the ambient gases, with effective orders β fitted at 1.18 to 1.45; the deck is a closed equilibrium titration with exact stoichiometric order 1, whose log-ratio is driven to $\pm\infty$ by cooling. Plotted against $\varepsilon = \beta \ln(C/C_{1/2})$, with occupancy $(S - S_{lo})/(S_{hi} - S_{lo})$, the binned readings of S1, S2, S3 and S5 collapse onto the single logistic $1/(1 + e^{-\varepsilon})$ on which the four planets sit at their NH₄SH bases [18].

8 The pair behind the periodic table

In his treatment of Döbereiner's triads [1] in atomic number, Scerri compared the medium-long and left-step (Janet) forms of the table and proposed placing hydrogen above fluorine to complete the triad H–F–Cl [2, 3, 4]. Comparing the two forms, he observed that the medium-long form is anomalous for groups 1 and 2 whereas the left-step form shows no exceptions [3], and that in the left-step form triads consistently align with periods of equal length [4]. We give the arithmetic behind those observations. The pair appears twice: as the spin pair that fixes every capacity (this section), and as the pair of Z -gaps around the middle member of a triad (Section 9). The counts below come from exhaustive enumeration of the 118-element table generated from the Madelung order, and the property values from arithmetic on tabulated atomic data; every input is listed in Section 13.

An orbital (n, ℓ, m_ℓ) holds a spin pair, so a subshell holds $2(2\ell + 1)$ electrons and a shell $\sum_{\ell < n} 2(2\ell + 1) = 2n^2$. The Madelung rule orders subshells by $n + \ell$ and, within equal $n + \ell$, by n [5]. It is an idealisation (several ground configurations, such as those of Cr and Cu, depart from it), but the block and group structure of the table follows it. Sorting the subshells (n, ℓ) , $0 \leq \ell < n$, by $(n + \ell, n)$ gives

$$1s \ 2s \ 2p \ 3s \ 3p \ 4s \ 3d \ 4p \ 5s \ 4d \ 5p \ 6s \ 4f \ 5d \ 6p \ 7s \ 5f \ 6d \ 7p.$$

Filling these subshells in turn with $Z = 1, \dots, 118$ and starting a new period at every ns subshell gives, by direct count, $L(1), \dots, L(7) = 2, 8, 8, 18, 18, 32, 32$ and period ends $\Sigma(p) = \sum_{q \leq p} L(q) = 2, 10, 18, 36, 54, 86, 118$.

Proposition 6 (period-pair law). *Let $\kappa(m)$ be the capacity of all subshells with $n + \ell = m$. Then $\kappa(m) = 2\lceil m/2 \rceil^2$, and period p consists of the ps subshell together with the subshells with $n + \ell = p + 1$ and $\ell \geq 1$, so*

$$L(p) = \kappa(p + 1) = 2 \left\lceil \frac{p+1}{2} \right\rceil^2.$$

In particular $L(2k) = L(2k + 1) = 2(k + 1)^2$ for $k \geq 1$, and $L(1) = 2$ is the only length that occurs once.

Proof. With $n = m - \ell$ and $\ell \leq n - 1$, the admissible ℓ are $0, 1, \dots, \lfloor (m - 1)/2 \rfloor$, so $\kappa(m) = 2 \sum_{\ell=0}^{\lfloor (m-1)/2 \rfloor} (2\ell + 1) = 2(\lfloor (m - 1)/2 \rfloor + 1)^2 = 2\lceil m/2 \rceil^2$. The admissible range of ℓ is the same for $m = 2k - 1$ and $m = 2k$, hence $\kappa(2k - 1) = \kappa(2k)$. Within $n + \ell = m$ the rule orders subshells by increasing n , so the s subshell ($n = m$) comes last. A period therefore extends from the ps subshell (the last member of group $m = p$) through the subshells of group $m = p + 1$ except its final $(p + 1)s$: one s subshell is exchanged for another, and $L(p) = \kappa(p + 1)$. Then $L(2k) = \kappa(2k + 1) = \kappa(2k + 2) = L(2k + 1)$. $\square$

Reading the periods off the order displayed above confirms the subshell content of every period and $L(p) = \kappa(p + 1)$ for $p = 1, \dots, 7$; the capacities are $\kappa(1), \dots, \kappa(9) = 2, 2, 8, 8, 18, 18, 32, 32, 50$. Period 1 is unpaired because its partner would be period 0. The left-step table, whose rows are the groups $n + \ell = m$, has row lengths $2, 2, 8, 8, 18, 18, 32, 32$ and no unpaired row; it is the form whose triads Scerri discusses [3, 4]. The same formula gives $L(8) = L(9) = 50$ for the Madelung order continued past $Z = 118$.

9 Exact triads

Convention. IUPAC group numbers 1–18, with H in group 1 and He in group 18. Group 3 is Sc, Y, Lu, Lr; La–Yb and Ac–No form the f-block and carry no group number. This is the assignment the Madelung order itself produces; the composition of group 3 is the subject of an IUPAC project, whose provisional report is [27]. Placing each element of the generated table by the rule stated after Theorem 7 (groups 1–2 counted from the left of the period, groups 3–18 from the right, the 28 f-block elements without a group) reproduces the IUPAC columns for all 18 groups; group 3 as Sc, Y, La, Ac is examined as a sensitivity check. A vertical triad is three consecutive members $Z_1 < Z_2 < Z_3$ of a group; it is exact if $Z_2 - Z_1 = Z_3 - Z_2$. The $118 - 28 = 90$ elements with a group number form $90 - 18 = 72$ pairs of consecutive members and $90 - 2 \cdot 18 = 54$ vertical triads. Every group has exactly one member in each period from its first period onwards, so consecutive members lie in consecutive periods.

With $\Sigma(p)$ the last atomic number of period p ($\Sigma(0) = 0$), for Z in period p let $\pi_l(Z) = Z - \Sigma(p - 1)$ be its place counted from the left and $\pi_r(Z) = \Sigma(p) - Z$ its place counted from the right.

Lemma 7 (alignment). *If Z lies in period p and Z' in period $p + 1$, then*

$$Z' - Z = L(p) + \pi_l(Z') - \pi_l(Z) = L(p + 1) - (\pi_r(Z') - \pi_r(Z)).$$

Hence $Z' - Z = L(p)$ if and only if $\pi_l(Z') = \pi_l(Z)$, and $Z' - Z = L(p + 1)$ if and only if $\pi_r(Z') = \pi_r(Z)$.

Proof. $Z' = \Sigma(p) + \pi_l(Z') = \Sigma(p + 1) - \pi_r(Z')$ and $Z = \Sigma(p - 1) + \pi_l(Z) = \Sigma(p) - \pi_r(Z)$; subtract. $\square$

The s-block opens every period, so groups 1 and 2 have $\pi_l = g$, with g the group number; the p-block closes every period and, in periods 4–7, the d-block precedes it immediately, so groups 3–18 have $\pi_r = 18 - g$ (He, at the end of period 1, has $\pi_r = 0$). Enumerating all 72 consecutive pairs confirms this: in groups 1–2 every pair has $\pi_l(Z') = \pi_l(Z)$ (left-aligned), in groups 3–18 every pair has $\pi_r(Z') = \pi_r(Z)$ (right-aligned), and no pair is neither.

Corollary 8. *For consecutive members $Z_1 < Z_2$ of a group, with Z_1 in period p_1 : in groups 1–2, $Z_2 - Z_1 = L(p_1)$, the length of the period of Z_1 ; in groups 3–18, $Z_2 - Z_1 = L(p_1 + 1)$, the length of the period of Z_2 . Outside the s-block the gap equals the length of the period of Z_1 exactly when $L(p_1) = L(p_1 + 1)$, that is when $p_1 \in \{2, 4, 6\}$ opens a pair of periods.*

Proof. Z_2 lies in period $p_1 + 1$. In groups 1–2, $\pi_l(Z_2) = \pi_l(Z_1) = g$, and Theorem 7 gives $Z_2 - Z_1 = L(p_1)$; in groups 3–18, $\pi_r(Z_2) = \pi_r(Z_1) = 18 - g$, and it gives $Z_2 - Z_1 = L(p_1 + 1)$. By Theorem 6, $L(p_1) = L(p_1 + 1)$ with $1 \leq p_1 \leq 6$ holds exactly for $p_1 = 2, 4, 6$. $\square$

Of the 72 pairs, 49 have $Z_2 - Z_1 = L(p_1)$, 67 have $Z_2 - Z_1 = L(p_2)$ and 44 both (direct enumeration). These counts also follow by hand: the 11 pairs of groups 1–2 all have $Z_2 - Z_1 = L(p_1)$, and 6 of them (those with $p_1 \in \{2, 4, 6\}$) also $L(p_2)$; the 61 pairs of groups 3–18 all have $Z_2 - Z_1 = L(p_2)$, and 38 of them (those with $p_1 \in \{2, 4, 6\}$) also $L(p_1)$; hence $11 + 38 = 49$, $61 + 6 = 67$ and $6 + 38 = 44$; none outside groups 1–2 has $Z_2 - Z_1 = L(p_1) \neq L(p_2)$. Thus Li→Na is $8 = L(2)$, but Cl→Br is $18 = L(4)$, not $L(3)$. This is the arithmetic form of the asymmetry between groups 1–2 and the rest noted by Scerri [3].

Theorem 9 (exact triads). *A vertical triad with members in periods $p, p + 1, p + 2$ is exact if and only if the two periods it spans have equal length: $L(p) = L(p + 1)$ in groups 1–2 and $L(p + 1) = L(p + 2)$ in groups 3–18. By Theorem 6 this holds if and only if p is even in groups 1–2 and odd in groups 3–18, and then the common gap s is the repeated length, $s \in \{8, 18, 32\}$.*

Proof. By Theorem 8 the two gaps are $L(p), L(p + 1)$ in groups 1–2 and $L(p + 1), L(p + 2)$ in groups 3–18. $\square$

Enumeration. The table has 54 vertical triads; 27 are exact, exactly those spanning equal-length periods. The count follows from Theorem 9, with p the period of Z_1 : groups 1 and 2 have triads starting at $p = 1, \dots, 5$ and $p = 2, \dots, 5$, of which $p = 2, 4$ are exact ($2 + 2$); groups 3–12 start at $p = 4, 5$, of which $p = 5$ is exact (10); groups 13–17 start at $p = 2, \dots, 5$, of which $p = 3, 5$ are exact (10); group 18 starts at $p = 1, \dots, 5$, of which $p = 1, 3, 5$ are exact (3); in all 27. By gap pair $(Z_2 - Z_1, Z_3 - Z_2)$ there are 3 triads at (8, 8), 8 at (18, 18) and 16 at (32, 32), against 1 at (2, 8), 8 at (8, 18) and 18 at (18, 32) (Fig. 3b). The exact triads are, for $s = 8$: Li–Na–K, Be–Mg–Ca, He–Ne–Ar; for $s = 18$: K–Rb–Cs, Ca–Sr–Ba, Al–Ga–In, Si–Ge–Sn, P–As–Sb, S–Se–Te, Cl–Br–I, Ar–Kr–Xe; for $s = 32$: the ten d-block triads Y–Lu–Lr, Zr–Hf–Rf, $\dots$, Cd–Hg–Cn (groups 3–12), and In–Tl–Nh, Sn–Pb–Fl, Sb–Bi–Mc, Te–Po–Lv, I–At–Ts, Xe–Rn–Og. Döbereiner’s four triads are all exact: Li–Na–K (3, 11, 19) with $s = 8$, and Ca–Sr–Ba (20, 38, 56), Cl–Br–I (17, 35, 53) and S–Se–Te (16, 34, 52) with $s = 18$.

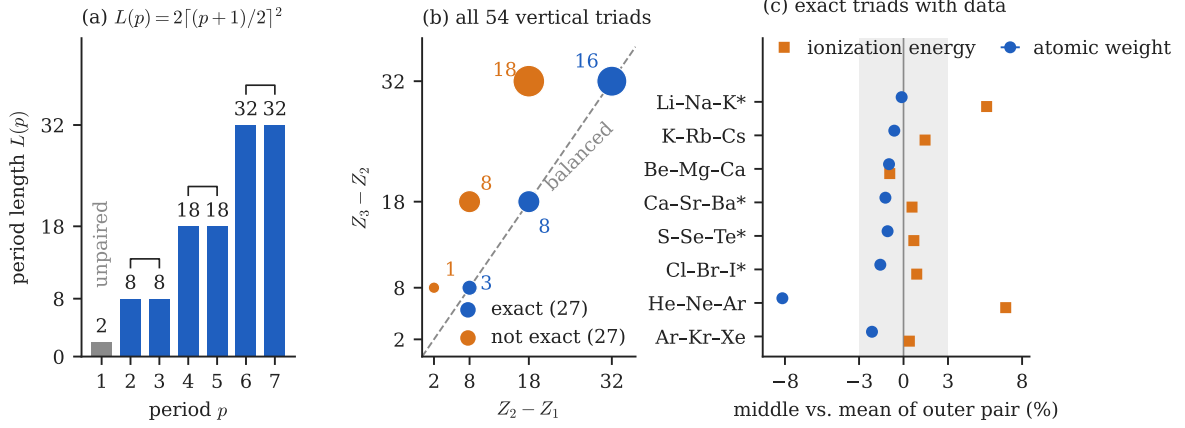


Figure 3: (a) Period lengths from the Madelung order; brackets join the equal pairs, and period 1 is unpaired. (b) The two Z -gaps of all 54 vertical triads; marker area grows with the number of triads (printed). Exact triads lie on the diagonal, at the repeated period lengths 8, 18, 32. (c) Deviation of the middle member from the mean of the outer pair for the exact triads in groups 1, 2, 16, 17, 18 with data: standard atomic weight (circles; bars from the abridged uncertainties) and first ionization energy (squares). * marks Döbereiner's triads; the band is $\pm 3\%$.

Pair, merge and observer. In an exact triad the middle element is the zero of the pair $(Z_1 - Z_2, Z_3 - Z_2) = (-s, +s)$, and the triad is $Z_2 + \{-s, 0, s\}$. Theorem 9 says when such a triad exists: two periods of equal length must follow each other, which is the period-pair law seen from inside a group. The gaps down a group are consecutive period lengths (group 1: 2, 8, 8, 18, 18, 32; group 18: 8, 8, 18, 18, 32, 32), so exact triads in one group meet end to end: K ends Li-Na-K and begins K-Rb-Cs; Ca, In, Sn, Sb, Te and I do the same in groups 2 and 13–17, and Ar and Xe link three exact triads in group 18. The shared end is the observer of the first triad and a member of the next: the chain is the merge of triads along a group.

Sensitivity and left-step form. With group 3 as Sc, Y, La, Ac, group 3 becomes left-aligned; there are again 54 triads, 27 exact and no mismatch, with Sc-Y-La (18, 18) exact in place of Y-Lu-Lr. In the left-step table (rows $n + \ell = m$, each row ordered f, d, p, s from left to right, so that He heads the s-block column above Be) every column is right-aligned; it has 54 triads, 26 exact, and exactness again coincides with equal row lengths. It loses He-Ne-Ar, since He sits above Be and He-Be-Mg has gaps (2, 8). Moving H above F, as Scerri proposed [2], adds the exact triad H-F-Cl (8, 8) and removes H-Li-Na, giving 27 of 54.

10 How far properties follow the balance

For a property x let $\eta = x_2/\bar{x} - 1$, $\bar{x} = \frac{1}{2}(x_1 + x_3)$, be the deviation of the middle member from the mean of the outer pair. Standard atomic weights are the CIAAW abridged values [26, 28]; treating the printed uncertainties u_i as independent standard uncertainties, first-order propagation gives

$$u_\eta^2 = \left(\frac{u_2}{\bar{x}}\right)^2 + \left(\frac{x_2}{2\bar{x}^2}\right)^2 (u_1^2 + u_3^2),$$

since $\partial\eta/\partial x_2 = 1/\bar{x}$ and $\partial\eta/\partial x_1 = \partial\eta/\partial x_3 = -x_2/(2\bar{x}^2)$. First ionization energies are those of the NIST Atomic Spectra Database [20], omitting values that the database marks as theoretical or estimated (printed in parentheses or brackets). All values used are listed in Table 16. Te-Po-Lv, I-At-Ts and Xe-Rn-Og lack the data, so eight exact triads remain (Table 15, Fig. 3c).

Table 15: Deviation of the middle member from the mean of the outer pair, exact triads with data. *: Döbereiner’s triads.

triad	group, s	η_A (%)	η_{IE} (%)	IE between
Li–Na–K*	1, 8	-0.13 ± 0.13	+5.61	yes
K–Rb–Cs	1, 18	-0.62 ± 0.01	+1.45	yes
Be–Mg–Ca	2, 8	-0.98 ± 0.01	−0.93	yes
Ca–Sr–Ba*	2, 18	-1.22 ± 0.01	+0.57	yes
S–Se–Te*	16, 18	-1.08 ± 0.02	+0.70	yes
Cl–Br–I*	17, 18	-1.57 ± 0.01	+0.89	yes
He–Ne–Ar	18, 8	-8.17 ± 0.33	+6.90	yes
Ar–Kr–Xe	18, 18	-2.13 ± 0.09	+0.39	yes

Atomic weight. The middle weight lies below the mean in all eight triads; for Döbereiner’s four, by 0.13 to 1.57%. With $w_i = A_i/Z_i$ and Z_2 the exact mean, $Z_1 = Z_2 - s$ and $Z_3 = Z_2 + s$, so $2A_2 - (A_1 + A_3) = 2w_2Z_2 - w_1(Z_2 - s) - w_3(Z_2 + s)$, which is the identity

$$2A_2 - (A_1 + A_3) = -Z_2(w_1 + w_3 - 2w_2) - s(w_3 - w_1).$$

In the $s = 18$ triads the second (trend) term, from the growth of w with Z (for example 2.058, 2.310, 2.417 along K–Rb–Cs), lies between -8.10 and -3.81 and outweighs the curvature term. The largest deviation, He–Ne–Ar, reflects the weight of argon, dominated by ^{40}Ar from the electron-capture branch of ^{40}K (Section 4) [29], which also places it above potassium (39.95 against 39.098). Where the Z -pair is not balanced, neither is the weight: for the five non-exact triads in these groups with weights and no hydrogen (Na–K–Rb, Mg–Ca–Sr, O–S–Se, F–Cl–Br, Ne–Ar–Kr) the middle Z lies 20.0 to 23.8% below the outer mean and the middle weight 23.2 to 32.5% below.

Ionization energy. In all six exact triads of groups 1, 2, 16 and 17 and in both group-18 triads the middle energy lies strictly between the outer two. In seven of the eight it lies above the mean. Without a member from the first two periods the deviation is at most 1.45% (K–Rb–Cs); with one it is +5.61% for Li–Na–K and +6.90% for He–Ne–Ar, and -0.93% for Be–Mg–Ca. The balance of Z carries over to the ionization energy most closely below the second period, and least well at the top of the table, where the first member of a group differs most from its congeners: the first-member anomaly that Scerri weighs in his case for the left-step table [4].

Where the balance breaks. *d-block contraction.* Ga, Ge, As, Se, Br and Kr follow the first filled 3d shell, whose electrons shield the nucleus poorly [29]. In all six exact triads with a 4p middle member, $\eta_{IE} > 0$ (+1.92, +1.96, +2.52, +0.70, +0.89, +0.39% for groups 13–18). For Ga the effect lifts its energy above that of Al, by 0.0135 eV, so Al–Ga–In, exact in Z , is the one exact triad with data whose middle energy is not between the outer two. *Relativistic effects.* In the non-exact triads Cu–Ag–Au and Zn–Cd–Hg the heaviest member has the highest ionization energy (Au exceeds Cu by 1.4992 eV, Hg exceeds Zn by 1.0433 eV), and in Rb–Cs–Fr and Sr–Ba–Ra the period-7 member lies above the period-6 member (4.0727 against 3.8939 eV, 5.2784 against 5.2117 eV). This is the pattern of the relativistic stabilisation of s shells in the heaviest elements [30]. The exact triads Ag–Au–Rg and Cd–Hg–Cn lack measured energies for their heaviest members; Y–Lu–Lr, with 4.96(5) eV for Lr, gives -2.91% .

11 Horizontal triads and lanthanide-contraction pairs

Horizontal triads have $s = 1$ and are exact in Z by construction. Fe–Co–Ni gives $\eta_A = +2.91\%$, because cobalt (58.933) is heavier than nickel (58.693), and $\eta_{IE} = +1.41\%$. Ru–Rh–Pd gives -0.80% and -4.97% , and Os–Ir–Pt gives -0.23% and $+3.09\%$, with Ir above both neighbours.

For Ti–Zr–Hf and V–Nb–Ta the Z -gaps are (18, 32): the second gap exceeds the first by $14 = 2(2 \cdot 3 + 1)$, the capacity of the 4f subshell. The Shannon ionic radii (six-coordinate) are 0.605, 0.72, 0.71 Å for Ti^{4+} , Zr^{4+} , Hf^{4+} and 0.54, 0.64, 0.64 Å for V^{5+} , Nb^{5+} , Ta^{5+} [31], as transcribed in [32]. The heavier two form a near-equal pair (ratio 0.986 and 1.000) although their weights differ by a factor of 1.957 and 1.948. This is the lanthanide contraction [29]. In these terms the radius triad collapses into a pair: one step from the first member and almost none between the heavier two, although in Z the second step is the longer.

12 Concluding remarks

In each case the chemically meaningful quantity is the difference of the pair: which configuration holds the electron, the net charge change of a branching, which level is the majority, whether two channels co-vary, which member holds a sensor surface’s electron, and which element is the zero of a triad. The balance is unreachable (Na/K), unrealised, with a difference that two evaluations place 2.6 combined standard uncertainties apart (^{40}K), crossed at a finite temperature fixed by degeneracy (Al, 232.61 K), manufactured by a shared code (absent measurements, where one coded correlation bounds how far the code sits from the data and the present-row moments reproduce it to 10^{-16}), crossed at a concentration fixed by the equality of release and trapping (the sensor ladder, from 1.68 to 7.49 mg m $^{-3}$ of CO), or exact by counting (the 27 triads that span equal-length periods). The same law decides the gate of a membrane channel [17] and the top ice deck of a giant planet [18].

Outlook. A record in which the oxidizing gas varies independently of the reducing ones, or one with an ozone reference, will place the oxidizing member and S5 on the same ladder and read the balance line of Section 7.2 directly. The half-point analysis applies unchanged to any array of metal-oxide sensors recorded beside reference analysers.

13 Data and sources

All numbers not listed here are computed from these inputs by the formulas and methods stated in the text.

- *Physical constants* (CODATA 2018 [22]): $k_B = 8.617333262 \times 10^{-5}$ eV K $^{-1}$ (exact in the SI); second radiation constant $hc/k_B = 1.438776877$ cm K; electron mass $m_e = 5.48579909 \times 10^{-4}$ u (used only for the 1.5×10^{-5} translational factor, with the mass numbers 23 and 39 for Na and K); molar gas constant $R = 8.314462618$ J mol $^{-1}$ K $^{-1}$ (for the NO $_2$ conversion factor).
- *Atomic data* (NIST Atomic Spectra Database, version 5.12 [20], accessed 24 September 2026; level designations as in [21]): first ionisation energies IE(Na I) = 5.13907696(25) eV and IE(K I) = 4.34066373(9) eV; ground levels Na $3s^2S_{1/2}$, K $4s^2S_{1/2}$, Na $^+$ and K $^+$ 1S_0 ; K I $4p^2P_{1/2}^\circ$ and $^2P_{3/2}^\circ$ at 12985.185724 and 13042.896027 cm $^{-1}$; Al I $3s^23p^2P_{1/2}^\circ$ at 0 and $^2P_{3/2}^\circ$ at 112.061 cm $^{-1}$, next level $3s^24s^2S_{1/2}$ at 25347.756 cm $^{-1}$. Degeneracies are $g = 2J + 1$. The first ionization energies of the triad sections are listed in Table 16.
- *Nuclear data for ^{40}K* : isotopic abundance 0.000117(1) [23]; ENSDF evaluation [6]: β^- 89.28(11)%, electron capture plus β^+ 10.72(11)%, β^+ 0.001%; KDK re-evaluation [7] (Ta-

ble IV there, with the partial decay constants of [8]): β^- 89.59(5)%, capture to the excited state of ^{40}Ar 10.31(4)%, capture to the ground state 0.098 ± 0.023 (stat) ± 0.010 (syst)%.

- *Standard atomic weights* A_r (Table 16): CIAAW abridged standard atomic weights 2024 [26], which are those of the 2021 report [28] with the 2024 revisions for Gd, Lu and Zr; accessed 24 September 2026. Fr, Ra and Lr have no standard atomic weight, and the period-7 elements from Rg to Og have neither quantity. Numbers in parentheses are the printed uncertainties in units of the last digit. The molar mass of NO_2 , $46.005 \text{ g mol}^{-1}$, is formed from the abridged weights of N (14.007) and O (15.999).
- *Structural inputs* for Sections 8 and 9: the capacities $2(2\ell + 1)$ and the Madelung $(n + \ell, n)$ order [5]; atomic numbers $Z = 1, \dots, 118$; IUPAC group numbers 1–18 with H in group 1, He in group 18 and group 3 as Sc, Y, Lu, Lr [27]; Döbereiner’s four triads Li–Na–K, Ca–Sr–Ba, Cl–Br–I, S–Se–Te [1, 2]. Everything in Sections 8 and 9 is computed from these by counting.
- *Ionic radii* (six-coordinate, Å): Ti^{4+} 0.605, Zr^{4+} 0.72, Hf^{4+} 0.71, V^{5+} 0.54, Nb^{5+} 0.64, Ta^{5+} 0.64, from Table 1 of [31], as transcribed in [32].
- *Dataset* (Sections 6 and 7): the UCI Air Quality data set [13], doi:10.24432/C59K5F, data-set page accessed 24 September 2026, introduced in [12]: hourly averaged responses of an array of five metal-oxide chemical sensors (PT08.S1 to PT08.S5) with reference-analyser concentrations and meteorological readings, recorded at road level in a polluted area of an Italian city, with missing values tagged -200 . Part used: the spreadsheet workbook distributed with the data set, all 9357 dated rows (10 March 2004 18:00 to 4 April 2005 14:00), columns PT08.S1 to PT08.S5, CO(GT), NMHC(GT), C6H6(GT) (shown in Section 7.9 to be a transform of PT08.S2), NOx(GT), NO2(GT), T, RH and AH. The sensor identities, units and the absent-value code are those stated on the data-set page. Every statistic of Sections 6 and 7 is computed directly from this workbook; the semicolon-separated text version of the data set, which rounds the sensor channels to integers, is used only for the comparison stated in Section 6.
- *Planetary N/S ratios* (Section 7.11): the nominal deep mole-fraction ratios 4.73, 3.16, 0.195 and 0.170 for Jupiter, Saturn, Uranus and Neptune, from the companion paper [18], which lists their observational sources.

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Table 16: Atomic data used in Sections 10 and 11: standard atomic weights A_r [26] and first ionization energies IE [20].

El.	Z	A_r	IE (eV)	El.	Z	A_r	IE (eV)
He	2	4.0026(1)	24.587389011(25)	Sr	38	87.62(1)	5.69486745(12)
Li	3	6.94(6)	5.391714996(22)	Y	39	88.906(1)	6.21726(10)
Be	4	9.0122(1)	9.322699(7)	Zr	40	91.222(3)	6.634126(5)
O	8	15.999(1)	13.618055(7)	Nb	41	92.906(1)	6.75885(4)
F	9	18.998(1)	17.42282(5)	Ru	44	101.07(2)	7.36050(5)
Ne	10	20.180(1)	21.564541(7)	Rh	45	102.91(1)	7.45890(5)
Na	11	22.990(1)	5.13907696(25)	Pd	46	106.42(1)	8.336839(10)
Mg	12	24.305(2)	7.646236(4)	Ag	47	107.87(1)	7.576234(25)
Al	13	26.982(1)	5.985769(3)	Cd	48	112.41(1)	8.993820(16)
Si	14	28.085(1)	8.15168(3)	In	49	114.82(1)	5.7863558(5)
P	15	30.974(1)	10.486686(15)	Sn	50	118.71(1)	7.343918(12)
S	16	32.06(2)	10.3600167(14)	Sb	51	121.76(1)	8.608389(12)
Cl	17	35.45(1)	12.967633(16)	Te	52	127.60(3)	9.009808(6)
Ar	18	39.95(16)	15.7596119(5)	I	53	126.90(1)	10.451236(25)
K	19	39.098(1)	4.34066373(9)	Xe	54	131.29(1)	12.1298437(15)
Ca	20	40.078(4)	6.1131549210(5)	Cs	55	132.91(1)	3.89390572743(17)
Fe	26	55.845(2)	7.9024681(12)	Ba	56	137.33(1)	5.2116646(12)
Co	27	58.933(1)	7.88101(12)	Lu	71	174.97(1)	5.425871(12)
Ni	28	58.693(1)	7.639878(17)	Hf	72	178.49(1)	6.825070(12)
Cu	29	63.546(3)	7.726380(4)	Ta	73	180.95(1)	7.549571(25)
Zn	30	65.38(2)	9.394197(6)	Os	76	190.23(3)	8.43823(20)
Ga	31	69.723(1)	5.9993020(12)	Ir	77	192.22(1)	8.96702(22)
Ge	32	72.630(8)	7.899435(12)	Pt	78	195.08(2)	8.95883(10)
As	33	74.922(1)	9.78855(25)	Au	79	196.97(1)	9.225554(4)
Se	34	78.971(8)	9.752368(6)	Hg	80	200.59(1)	10.437504(6)
Br	35	79.904(3)	11.81381(6)	Fr	87	–	4.0727411(11)
Kr	36	83.798(2)	13.9996055(20)	Ra	88	–	5.2784239(25)
Rb	37	85.468(1)	4.1771281(12)	Lr	103	–	4.96(5)

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